

Saving Responses to Mandatory Pension Plans*

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Abstract

To boost retirement savings, many countries mandate worker contributions to pension accounts. This paper investigates saving responses to such mandates throughout the entire portfolio, leveraging detailed administrative tax data from Switzerland and a regression discontinuity design. I find that mandatory pension plans have limited effects on total savings, with an estimated crowd-out rate of 94%. Decomposing the saving response, I show that workers offset mandatory pension contributions by reducing private non-retirement savings, primarily in financial assets. By contrast, there is no substitution between mandatory and voluntary pension savings. Liquidity-constrained workers are less able to reduce private savings in response to mandatory contributions and therefore increase their total savings.

JEL codes: D14, G51, H24, H31, J32

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1 Introduction

Population aging is putting pressure on pension systems, making it increasingly difficult to provide adequate financial support to retirees (Poterba, 2014). In response, many governments have implemented policies aimed at boosting retirement savings during individuals' working lives, including mandates that require contributions to pension accounts. As of 2022, 19 of the 38 OECD countries had adopted some form of mandatory pension plans (OECD, 2023). Even in countries without a comprehensive mandate, such as the US, some employer-sponsored pension plans feature mandatory contributions (Card and Ransom, 2011; Friedberg, Leive and Cai, 2024). Typically, these mandates require workers to contribute a fixed share of their earnings, with contributions withheld at source by employers and deposited directly into illiquid pension accounts. This facilitates the accumulation of retirement savings but reduces workers' disposable income.

Do pension plan mandates succeed in boosting savings? Ex ante, this is unclear. Individuals may respond by reducing other types of savings or taking on additional debt to maintain the same savings level as in the absence of the mandate. In such cases, mandatory contributions crowd out other savings, altering the portfolio composition without changing overall savings levels. Therefore, it is key to analyze saving responses across the entire portfolio to understand how mandatory pension plans affect wealth accumulation and financial preparedness for retirement. However, there is limited prior evidence due to both data constraints and identification challenges. Measuring savings across all asset classes is difficult, as administrative microdata usually lack comprehensive information on household savings and wealth. Moreover, credible identification of saving responses requires exogenous variation in mandate coverage.

This paper estimates the effect of mandatory pension plans on overall savings and characterizes the saving response throughout the portfolio, leveraging administrative tax data from the Swiss canton of Bern and identifying variation from the occupational pension plan mandate. Detailed information on savings and wealth allows me to decompose the response by asset type, including financial wealth, business wealth, property, other wealth, debt, and voluntary pension savings. Further, I examine how liquidity constraints influence workers' saving responses.

Switzerland is a particularly suitable laboratory for studying saving responses to mandatory pension plans as the rich administrative data and institutional context enable me to overcome the obstacles highlighted above. I draw on income tax and wealth tax records to construct comprehensive measures of savings at the asset-type level. The universal coverage of the administrative data ensures representativeness and provides ample statistical power to precisely estimate effects. Measurement er-

ror and misreporting, common concerns with survey data, are limited because the tax authority verifies the tax records for administrative purposes. With savings data in hand, I approximate the ideal experiment exploiting the fact that the pension plan mandate only applies to workers with earnings above a threshold, providing compelling identifying variation in mandatory retirement contributions among otherwise very similar workers. The institutional setting in Switzerland is comparable to many other high- and middle-income countries. The pension system consists of three pillars: a pay-as-you-go scheme, employer-based pension plans, and tax-advantaged private pension accounts. This increases the likelihood that the findings of this paper are generalizable to other contexts.

The Swiss pension plan mandate requires employees whose annual earnings exceed a threshold of around CHF 20,000 (corresponding roughly to the 30th percentile of the earnings distribution) to contribute to an employer-sponsored fully funded pension account.¹ Individuals subject to the mandate cannot opt out. Contributions are calculated by applying a legally defined age-specific contribution rate to qualifying earnings which equal gross earnings minus a deduction. For workers whose earnings are marginally above the mandate threshold, the contribution rate is applied to a fixed minimum amount. This causes a discontinuity in mandatory pension savings at the cutoff, meaning that otherwise similar workers are required to contribute substantially different amounts to employer-based pension plans. In contrast to many public pension systems, these mandatory pension plans feature a strong and salient link between contributions and future benefits. Because being enrolled in a mandatory pension plan raises the cap on contributions to preferentially taxed private pension accounts, there is also a discontinuity in the cap on private pension savings at the mandate threshold.

I combine the sharp policy variation with high-quality administrative tax data on income, wealth, and savings from the second-largest Swiss canton, Bern, spanning years 2005 to 2017. The data cover the entire adult population and provide detailed information on workers' voluntary retirement savings, wealth, and debt, enabling me to construct savings measures by asset type. They also allow me to link partners to include household-level variables in the analysis. These data are quite exceptional as Switzerland is one of only three OECD countries that still levy a comprehensive wealth tax (OECD, 2018c).

To estimate the saving responses to mandatory pension plans, I leverage the variation in mandate coverage provided by the earnings threshold within a regression discontinuity framework. As workers with earnings just below and just above the threshold are very similar in all dimensions, except that only those above are sub-

¹For most of the sample period 2005–2017, the Swiss franc (CHF) was trading roughly at parity with the US dollar. Therefore, I do not report separate figures in US dollar.

ject to the mandate, this allows me to identify the causal effect. I estimate that the mandate causes a discontinuity in mandatory retirement savings of CHF 406, corresponding to approximately 2% of workers' earnings.

Despite the sizeable mandatory pension contributions, I find that the mandate's effect on total savings is limited. The point estimate is CHF 26, implying a crowd-out rate of 94%. While the estimated effect on total savings is not statistically significantly different from zero, I can reject an increase by more than CHF 290 (or equivalently, crowd-out rates lower than 29%) at the 5% level. This implies that workers substantially offset mandatory retirement contributions by reducing other savings.

To identify the types of savings that drive the crowding-out response, I decompose the overall effect throughout the portfolio. Estimating separate effects for each asset type, I find that workers respond to the mandate by reducing private *non-pension* savings by CHF 459, which is of similar magnitude as the mandated pension contributions. The decline in private savings is mainly driven by a CHF 311 decrease in savings in financial assets, which include bank accounts, stocks, bonds, and other securities. I do not find statistically significant effects on other types of assets such as business wealth, property, other types of wealth, or debt, and the precision of most estimates allows me to rule out even modest changes. It seems intuitive that workers respond to the mandate primarily by lowering savings in financial assets as they are more liquid than other types of wealth.

While the mandate reduces private savings, I estimate that it increases voluntary pension savings by CHF 112. This positive effect appears surprising, but it reflects the combination of the potential impacts of mandatory pension savings and the cap on preferentially taxed private pension contributions, which both increase discontinuously at the threshold. I disentangle these two forces by (i) focusing on the lower end of the distribution of private pension savings where the contribution cap is irrelevant, (ii) imposing the same cap above and below the threshold, and (iii) assessing the mandate's effect on a component of voluntary pension savings – lump-sum buy-ins into occupational pension plans – that is not subject to the cap. I document clear evidence that the positive effect on voluntary pension savings is entirely driven by the increase in the contribution cap. After removing the impact of the cap, I find no substitution between mandatory and voluntary pension savings.

These results imply that mandatory pension plans have limited effects on total savings, but they induce a reallocation of savings from more liquid financial assets to less liquid pension accounts. However, average effects may mask important heterogeneity in saving responses. In particular, standard models of saving under borrowing constraints predict that liquidity-constrained workers are less able to decrease their private savings to offset mandatory pension contributions. To investigate the role of liquidity constraints, I estimate heterogeneous effects across workers with differ-

ent levels of baseline financial assets or household income, finding consistent results across these two proxies for liquidity. Consistent with theoretical predictions, the negative response of private savings to the mandate is substantially weaker among liquidity-constrained workers. Consequently, the mandate increases their total savings by around CHF 170–350, representing 1–2% of earnings. Workers with high levels of financial wealth or household income respond to the mandate by disproportionately lowering private savings in order to both offset mandatory pension contributions and redirect savings into preferentially taxed private pension accounts, taking advantage of the increased contribution cap. At the same time, the estimated effect on their total savings is close to zero. Whereas the regression discontinuity approach provides a local effect estimate for workers near the mandate threshold, the full crowding-out among those with high household income suggests that crowd-out rates further up the income distribution are likely at least as large as the 94% headline estimate.

This paper contributes to three strands of the literature on saving responses to pension policies. First, it adds to the thin evidence base on mandatory pension plans. While vast literatures study the savings effects of other types of pension policies such as financial incentives or automatic enrollment (see, e.g., [Choi, 2015](#); [OECD, 2018a](#)), we lack systematic evidence on the saving responses to mandatory pension plan contributions. The few existing papers find weak to moderate crowding-out of other savings based on data from Nordic countries or the US ([Card and Ransom, 2011](#); [Arnberg and Barslund, 2014](#); [Chetty et al., 2014](#); [Friedberg, Leive and Cai, 2024](#); [Jensen et al., 2025](#)). In a seminal paper, [Chetty et al. \(2014\)](#) use administrative data from Denmark and a regression discontinuity approach similar to this paper, finding little crowding-out. But the identifying variation in mandatory retirement savings is only around \$50 – an order of magnitude smaller than in the Swiss setting – and the threshold is at roughly \$5,000, so the marginal workers affected by the mandate have very low earnings and likely face liquidity constraints. [Card and Ransom \(2011\)](#) and [Friedberg, Leive and Cai \(2024\)](#) study the effect of mandatory pension contributions on voluntary retirement savings among faculty members at US universities. [Card and Ransom \(2011\)](#) find crowd-out rates of 60–80% for mandatory employee contributions, but only half as much for employer contributions. [Friedberg, Leive and Cai \(2024\)](#) estimate a lower crowd-out rate of 30%. Because they rely on data from pension plan providers, neither of the two papers accounts for saving responses outside pension accounts, resulting in a lower-bound estimate of the overall crowd-out rate.

In this paper, I estimate precise savings effects of mandatory pension contributions across all asset types, providing novel evidence of portfolio reallocation responses to pension policies. In particular, I find no substitution between mandatory and voluntary pension savings but strong crowding-out of non-pension savings, which is

not detected in prior work using pension plan data. In the theoretical framework of Chetty et al. (2014), these results are consistent with passive saving behavior under a consumption target: workers do not appear to re-optimize their voluntary pension savings, but instead hold consumption constant, so the decline in disposable income leads to crowding-out of non-pension savings.

In concurrent work, Jensen et al. (2025) find that a substantial increase in mandatory *employer* contributions for private-sector workers in Iceland did not crowd out voluntary savings. Two main factors likely explain the differences in our crowd-out estimates: (i) Jensen et al. (2025) examine a reform that raised only employer contributions and thus did not affect workers' disposable income, whereas the increase in contributions I study is split equally between employers and employees. As workers fund private savings out of disposable income, passive saving behavior is consistent with no crowding-out in response to employer contributions but significant crowding-out from employee contributions. (ii) I assess saving responses to an increase in mandatory contributions from a baseline of zero, coinciding with pension plan enrollment. This is likely more salient than the change in contribution rates studied by Jensen et al. (2025), which applied to workers already enrolled in a pension scheme and did not affect disposable income. In line with this interpretation, Jensen et al. (2025) find that most workers were unaware of the reform.

Second, my findings are related to the literature on automatic saving policies, such as automatic enrollment and defaults, which aim to nudge workers into raising retirement savings. The key difference between mandatory pension plans and automatic saving policies is that mandates do not allow individuals to opt out, which constrains the set of available behavioral responses. The evidence consistently shows that automatic saving policies substantially increase pension plan enrollment and contributions in the short run (Madrian and Shea, 2001; Choi et al., 2004; Thaler and Benartzi, 2004; Beshears et al., 2009; Benartzi and Thaler, 2013; Chetty et al., 2014; Blumenstock, Callen and Ghani, 2018; Cribb and Emmerson, 2020; Chalmers et al., 2022; Bucher-Koenen, Wallossek and Winter, 2024). But recent research documents that workers undo part of these savings gains in the medium to long term, e.g., by contributing at lower rates at subsequent employers without automatic enrollment and withdrawing funds upon job separation (Derby, Mackie and Mortenson, 2023; Falk and Karamcheva, 2023; Choi et al., 2024; Choukhmane, 2025). These findings highlight potential advantages of mandates over nudges, as mandates prohibit workers from opting out or decreasing contribution rates below the statutory minimum (as well as limiting withdrawals). My results complement this evidence, showing that even strict mandate policies may fail to boost overall savings as workers respond by reducing private savings outside retirement accounts. But mandatory pension plans can still improve financial preparedness for retirement by reallocating savings from

liquid private accounts to illiquid pension accounts which are more likely to remain untouched until retirement.

Whereas the earlier literature on auto-enrollment focused exclusively on retirement savings, recent work examines the effects on certain financial assets and debt. Using detailed data from a UK bank, [Choukhmane and Palmer \(2025\)](#) find that two-thirds of increased pension contributions are funded by reduced deposit balances and increased debt. This is consistent with [Beshears et al. \(2024\)](#) who show that employees of small UK businesses partly finance automatic savings with higher unsecured debt. By contrast, [Beshears et al. \(2022\)](#) do not find an effect on debt or credit scores of civilian employees of the US Army. I advance the emerging literature on the response of non-retirement savings to pension policies by comprehensively estimating effects for each asset type, including but not limited to financial wealth and debt.

Third, this paper relates to the longstanding literature on the savings effects of public pension systems, which also rely on mandatory contributions. The evidence suggests that private savings respond significantly to changes in public pension contributions and benefit eligibility, but the degree of substitution varies across settings and reforms ([Feldstein, 1974](#); [Attanasio and Brugiavini, 2003](#); [Attanasio and Rohwedder, 2003](#); [Bottazzi, Jappelli and Padula, 2006](#); [Aguila, 2011](#); [Lachowska and Myck, 2018](#); [Lindeboom and Montizaan, 2020](#); [Etgeton, Fischer and Ye, 2023](#); [García-Miralles and Leganza, 2024](#)). I show that mandatory pension plans, which are characterized by a stronger and more salient tax-benefit linkage than public pensions ([Bozio et al., 2025](#)), generate strong crowding-out effects. While the literature on public pensions typically relies on survey data, I use high-quality administrative data to estimate saving responses throughout the portfolio, similar to [García-Miralles and Leganza \(2024\)](#) who study the effect of delayed public pension benefit eligibility on private savings.

The remainder of this paper is organized as follows. In Section 2, I discuss the relevant parts of the Swiss pension system. Section 3 describes the administrative tax data and data preparation. Section 4 explains the regression discontinuity approach that exploits the discontinuity in mandate coverage at the earnings threshold. Section 5 presents the main results on the saving responses to mandatory pension plans. Section 6 explores the role of liquidity constraints. Section 7 concludes.

2 Institutional Background

The Swiss old-age provision system consists of three pillars: (i) a standard pay-as-you-go system, (ii) a fully funded occupational pension system, which includes the mandatory pension plans studied in this paper, and (iii) voluntary private pension accounts. The structure is similar to pension systems in many other countries that

consist of a government-backed defined-benefit system, employer-sponsored defined-contribution plans, and preferentially taxed private pension accounts (e.g., in Denmark or the US, see [Chetty et al., 2014](#)).² This section introduces the relevant institutional context, focusing on the rules of the occupational pension system and voluntary private pension accounts from 2005 to 2017, the years used in the main analysis. There have been no major changes to the pension system during this period.³ Appendix B describes the Swiss old-age provision system in more detail.

2.1 Occupational Pension System

The occupational pension system requires most workers to contribute to an employer-sponsored pension account. Employees must be enrolled in a pension plan by their employer if their annual gross earnings in the main job exceed a legally defined threshold. The pension savings mandate applies to women between 25 and 64 years and men between 25 and 65 years of age. The mandate is binding: employees meeting these conditions cannot opt out. Consequently, coverage is quite comprehensive. In 2017, 4.2 million individuals were enrolled in an occupational pension fund, representing around 83% of the Swiss labor force ([Federal Statistical Office, 2019](#)). Total contributions equalled CHF 54 billion, equivalent to 8% of GDP. These numbers highlight that occupational pension plans are one of the most important savings instruments in Switzerland.

The earnings threshold determining mandate coverage has increased gradually from CHF 19,350 in 2005 to CHF 21,150 in 2017.⁴ While the vast majority of occupational pension plans use the statutory threshold, some may enroll employees with earnings below. I cannot directly investigate this because occupational pension contributions are not observed in the tax data. But [Ecoplan \(2010\)](#) show that there are only small discrepancies between the number of employees enrolled in occupational pension plans according to information from plan providers and the corresponding counts computed using earnings data collected by the social insurance system. Thus, I can reliably infer pension plan enrollment from workers' earnings.

Mandatory occupational pension savings are calculated by applying a contribution rate to qualifying earnings which equal gross earnings minus a deduction. The deduction has increased from CHF 22,575 in 2005 to CHF 24,675 in 2017 and serves

²Simplifying, the corresponding three pillars in the US are Social Security, 401(k) plans, and IRAs.

³The occupational pension system was reformed between 2004 and 2006, but the relevant changes were implemented by 2005. Appendix Section B.4 provides more information about the reform.

⁴Like most parameters of the occupational pension system, the threshold is set as a function of the maximum pension in the pay-as-you-go system in order to avoid overinsurance. Benefit levels in the pay-as-you-go system are usually adjusted every other year based on an index capturing the mean of nominal wage growth and inflation. Note that pay-as-you-go contributions are levied on total earnings, without a cap, even if they do not lead to higher benefits.

a similar purpose as the mandate threshold. The statutory age-specific contribution rates are 7% for ages 25–34, 10% for ages 35–44, 15% for ages 45–54, and 18% for ages 55–64 among women and ages 55–65 among men. Employers must pay at least half the contribution. The share paid by employees is deducted from their earnings by the employer and directly transferred to the pension fund, along with the employer part. Contributions are exempt from income tax.

Importantly, there is a minimum for qualifying earnings. This has risen from CHF 3,225 to CHF 3,525 over the sample period. For employees whose earnings exceed the mandate threshold by only a small amount, the contribution rate is not applied to the marginal earnings above the threshold or deduction but to that minimum amount. This creates a discontinuity in mandatory pension contributions at the cutoff. I leverage this source of exogenous variation to study the effects of the mandate. Appendix Figure A.1 illustrates the calculation of mandatory pension contributions as a function of gross earnings for a worker aged 45–54 in 2017.⁵

In addition to mandatory contributions, many occupational pension plans allow lump-sum “buy-ins” which can also be deducted from taxable income.⁶ These buy-ins are directly observed in the tax records.

Workers are regularly informed about their contributions, investment returns, the capital in their pension account, and projected retirement benefits. Upon retirement, the accumulated pension wealth can be withdrawn in the form of a lifelong annuity, a lump-sum payment, or a combination of the two. This underscores the strong and salient link between mandatory pension contributions and future benefits. While contributions and returns on investment are exempt from income tax (and there is no capital gains tax in Switzerland) and occupational pension wealth is exempt from wealth tax, retirement benefits are taxed.⁷

2.2 Private Pension Savings

In addition to contributing to the pay-as-you-go system and occupational pension plans, employees can make voluntary contributions to designated private pension accounts. These contributions can also be deducted from taxable income up to an annual cap. The access to those funds is restricted until individuals enter retirement. Private pension accounts need to be set up separately with a bank or insurance company. In the remainder of this paper, I refer to this savings vehicle as “private pension savings”. Total private pension savings equal roughly CHF 10 billion per year, representing 1.5% of GDP (Schüpbach and Müller, 2019).

⁵Equation (4) in Appendix Section C.4 shows how to compute mandatory pension savings.

⁶Individuals can make buy-ins up to the level of savings that they would have accumulated, had they been earning their current salary since they were 25 years old (Kuhn, 2020).

⁷The majority of OECD countries applies a similar “exempt-exempt-taxed” regime (OECD, 2018b).

The contribution cap depends on whether workers are enrolled in an occupational pension plan. If they are enrolled, they can make yearly private pension contributions up to a fixed cap which has grown from CHF 6,192 in 2005 to CHF 6,768 in 2017. Workers not enrolled in an occupational pension plan can contribute up to 20% of their net earnings.⁸ Importantly, this means that the cap is significantly lower for workers just below the mandate threshold who are not enrolled in an occupational pension plan than for workers just above. Below but close to the threshold, the cap is between CHF 3,000 and CHF 4,000 in all years in the estimation sample.⁹ Workers above the cutoff are required to be enrolled in an occupational pension plan, so they can all make private pension contributions up to the fixed cap between CHF 6,000 and CHF 7,000. In the empirical analysis, I carefully disentangle the effect of mandatory pension contributions from the impact of the contribution cap, which both increase discontinuously at the mandate threshold.

There are strong tax incentives for private pension savings, in particular for high-income and high-wealth individuals, because contributions are deductible from taxable income and the accumulated capital is exempt from the wealth tax. Upon retirement, private pension capital can be received as an annuity subject to income tax or as a lump sum taxed at special but usually preferential rates.

3 Data

Assessing the saving responses to the pension plan mandate requires comprehensive individual-level measures of savings. These are challenging to obtain because high-quality microdata on assets and savings are scarce. Switzerland is a particularly well-suited setting to study saving choices and wealth accumulation due to the availability of administrative tax data on wealth.¹⁰ However, there is no individual-level dataset on wealth with nationwide coverage because the wealth tax is only collected at the cantonal and municipal level.

I draw on administrative tax microdata from the canton of Bern, which is the second-largest Swiss canton by population and approximately representative for Switzerland (Brunner, Meier and Näf, 2020).¹¹ The tax records provide detailed and

⁸There is also a fixed upper bound on private pension contributions for individuals not enrolled in an occupational plan which has increased from CHF 30,960 to CHF 33,840 over the sample period. This applies in particular to the self-employed, but it is not relevant for employees included in my analysis because the cap equal to 20% of net earnings binds first.

⁹The mandate threshold is around CHF 20,000 in gross earnings and the employee share of social insurance contributions a bit above 6%, with the exact numbers depending on the year. Therefore, the simplified calculation for the lower contribution cap is: $\text{CHF } 20,000 \times (1 - 0.06) \times 0.2 = \text{CHF } 3,760$. The cap can be higher if workers have additional earnings, e.g., from working a second job.

¹⁰Since France replaced its wealth tax with a tax on real estate in 2018, only two OECD countries other than Switzerland still levy a wealth tax: Norway and Spain (OECD, 2018c).

¹¹According to the Federal Statistical Office, 1.03 million people were resident in the canton of Bern

comprehensive information on income, wealth, and savings. They also contain basic demographic information including the year of birth, gender, marital status, the number of children, and the municipality of residence. The dataset is an annual panel covering all taxpayers in the canton between 2002 and 2017. It is based on the tax returns that all permanent residents must file. Accordingly, the composition of individuals in the data only changes due to migration or death. Full coverage of the population is key both for representativeness and to have enough statistical power when zooming in on the group of workers around the mandate threshold in the regression discontinuity analysis. As taxes are levied at the household level in Switzerland, I can link individuals to their partners if they are married or in a civil union. This allows me to include household-level variables in the analysis. The tax records are verified by the tax authority for administrative purposes, limiting the extent of measurement error and misreporting.

Because the pension plan mandate applies at the individual level while the tax unit in Switzerland is the household, I split up married couples into individual observations. Most of the important variables for the analysis, including earnings from work, private pension savings, and occupational pension buy-ins, are reported at the individual level in the tax data. For all income and wealth components reported only at the household level, I equally assign half to each partner (following [Fagereng et al., 2020](#)). Appendix C describes the dataset, data cleaning and preparation, and variable construction in more detail.

3.1 Income

The tax records provide detailed information on all types of income. This includes earnings from employment separately for the main job and other jobs, which is crucial for constructing the earnings measure determining whether workers are subject to the pension plan mandate. The data also provide information on self-employment income, business income, financial income, property income, transfer income, and pension income.

The relevant earnings concept for the pension plan mandate is gross earnings in the main job, before deducting social insurance and occupational pension contributions. This is the running variable in the regression discontinuity approach. Because I observe main-job *net* earnings in the tax data, I build a calculator using the year-specific social insurance and occupational pension contribution schedules to back out gross earnings for each individual. While this is straightforward for most individuals, gross earnings narrowly below and above the mandate threshold can result in

in 2017: <https://www.bfs.admin.ch/bfs/en/home/statistics/population.html> [accessed on 22 October 2021].

the same net earnings recorded in the tax data, because workers above the threshold have the employee part of mandatory pension contributions deducted whereas those below do not. Thus, for a small share of observations within a narrow range of CHF 350 around the threshold, I cannot unambiguously impute gross earnings. I exclude these observations from the main analysis.¹²

3.2 Wealth

The tax data also provide very comprehensive information on wealth which I use to construct savings measures by asset class. This includes financial wealth, business wealth, property wealth, other types of wealth, and debt. Financial wealth captures bank accounts, stocks, bonds, and any other types of financial investments. Business wealth reflects ownership of assets related to the operation of a trade or business, e.g., in self-employment, partnerships, or corporations. Property wealth includes any property people own, including primary residences, but the valuation is typically below the true market value and only updated periodically.¹³ “Other wealth” includes the value of an individual’s share of assets held through joint heirship or other forms of co-ownership and any assets not included elsewhere such as cash, gold, valuable art, cars, boats, horses, etc. Debt covers all types of private debt such as mortgages, loans, overdrafts, and credit card debt.

The only type of wealth missing from the data is pension wealth held in occupational and private pension accounts as these are exempt from the wealth tax. This is not a concern for the analysis as I can directly observe voluntary occupational and private pension savings in the tax data and impute mandatory occupational pension savings based on individuals’ gross earnings and the relevant contribution schedule.

3.3 Savings

To examine the effects of mandatory pension plans on the entire savings portfolio, I obtain measures of savings for each asset type. Whereas some savings measures are directly observable in the tax records, others can be computed based on available information.

Mandatory Pension Savings. Mandatory occupational pension contributions are not included in the data because they are withheld at source by the employer. I impute these by applying the statutory contribution schedule to gross earnings. I define

¹²This affects 16,607 individual-year observations, representing 2.6% of observations within the estimation window.

¹³As a rule of thumb, the tax value of property corresponds to around 60% of the market value (OECD, 2018c).

mandatory pension savings as the sum of employee and employer contributions, as both equally help build workers' pension account balances.

Voluntary Pension Savings. Voluntary pension savings include contributions to private pension accounts and lump-sum buy-ins into occupational pension plans. Both of these are observed in the tax data because they are deductible from taxable income.

Private Savings. For all types of non-pension savings, I construct annual savings measures using the wealth data. As is common in the literature (e.g., Chetty et al., 2014; García-Miralles and Leganza, 2024), I compute savings as the change in the reported value of assets in year t relative to year $t - 1$. Overall private savings are calculated as the change in net wealth relative to the previous year, which takes into account all types of non-pension assets and debt. These measures include both “active savings” and “passive savings” from price changes (see Fagereng et al., 2025). This is conceptually attractive as it is the most comprehensive savings measure in line with the classic Haig-Simons income concept (Aguiar, Moll and Scheuer, 2024). But in practice, it means that private savings are volatile due to fluctuations in asset prices.¹⁴ To reduce the variance stemming from asset price volatility, I follow García-Miralles and Leganza (2024) and winsorize overall private savings and asset-type-specific private savings at percentiles 5 and 95 of the distribution in the main estimation sample.¹⁵

When comparing private savings and pension savings, and constructing a consistent measure of total savings, I must account for the fact that mandatory and voluntary pension savings are tax-deductible and hence reported pre-tax, while private savings are measured post-tax. Intuitively, increasing pension savings by CHF 1 reduces consumption by only CHF $(1 - \tau_{it})$, where τ_{it} is the individual's marginal income tax rate, whereas increases in private savings reduce consumption one-for-one. Following Chetty et al. (2014), I compute pre-tax private savings by dividing observed post-tax private savings by the individual's net-of-marginal-tax rate.¹⁶

Total Savings. Total savings are defined as the sum of mandatory pension savings, voluntary pension savings, and overall private savings – all measured pre-tax.

¹⁴I cannot strip out asset price changes because data at the level of individual assets or transactions are not available.

¹⁵The exceptions are savings in business wealth and property wealth which I winsorize at percentiles 2.5 and 97.5 because they are rare for workers in the estimation sample. Only 9.5% and 7.9% of worker-year observations have non-zero business savings or property savings, respectively.

¹⁶Because the income tax schedule in the canton of Bern is smooth, with gradually changing marginal tax rates, I estimate marginal tax rates across the distribution by fitting a global fifth-order polynomial regression of the income tax liability on household income separately for singles and married couples.

4 Empirical Strategy

Estimating the causal effect of mandatory pension plans on savings is challenging because one needs to disentangle the saving response to the mandate from unobserved heterogeneity in saving preferences. The ideal, but impractical, experiment would force randomly selected individuals to contribute to pension accounts. I approximate this using a regression discontinuity design that leverages the earnings threshold of the Swiss pension plan mandate which requires otherwise similar workers to contribute substantially different amounts to employer-based pension accounts. The sharp mandate threshold provides credibly exogenous variation in mandate coverage that I can use to identify its causal effect. Combined with detailed and comprehensive tax records, this allows me to examine the saving response across the entire portfolio.

4.1 Discontinuity in Mandatory Pension Contributions

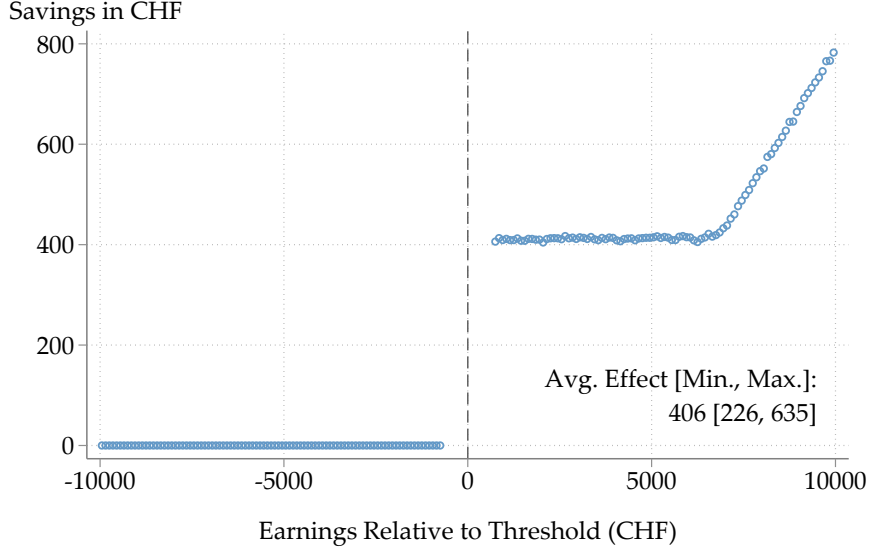
Employees with earnings above the mandate threshold automatically contribute a fraction of their salary to an occupational pension account, while those below are not required to make those savings. For those marginally above the threshold, the contribution rate is applied to the minimum qualifying earnings. Thus, annual mandatory pension savings jump from zero to CHF 225–635 at the cutoff, with the exact amount depending on the age-specific contribution rate and the year-specific minimum qualifying earnings. Figure 1 depicts the average mechanical effect of the mandate on mandatory pension savings. There is a discontinuity of CHF 406 at the threshold, corresponding to roughly 2% of earnings. I leverage this variation using a sharp regression discontinuity (RD) design to estimate the effect of mandatory pension plans on saving behavior.

4.2 Regression Discontinuity Design

Identification. Assuming that workers do not adjust their earnings to sort below or above the mandate threshold, the cutoff generates variation in treatment assignment that is orthogonal to other individual characteristics. The fundamental idea is that workers just below the threshold can be used as a valid counterfactual for those just above the threshold under the assumption that they are comparable in all relevant dimensions except for whether they are subject to the mandate.

Formally, identification requires that the (average) potential outcomes are continuous across the threshold (Hahn, Todd and Van der Klaauw, 2001). The continuity assumption can be expressed in the following way. Let $Y_{it}(1)$ and $Y_{it}(0)$ denote the savings outcome of interest for individual i in year t who is subject or not subject to the mandate, respectively. The sharp RD estimand β_{RD} identifies the local average

FIGURE 1: EFFECT OF PENSION PLAN MANDATE ON MANDATORY PENSION SAVINGS



Notes: Regression discontinuity plot showing the effect of the pension plan mandate on mandatory pension savings. Effective estimation sample includes 597,120 worker-year observations from 2005 to 2017. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff. The running variable is recentered around the mandate threshold, indicated by the dashed vertical line. Occupational pension savings are predicted by applying the statutory contribution schedule to gross earnings in the main job following Equation (4). See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

treatment effect on individuals with near-threshold earnings X_{it} , provided that the conditional expectation functions are continuous in x at the cutoff c_t :

$$\beta_{RD}(c_t) = \mathbb{E}[Y_{it}(1) - Y_{it}(0)|X_{it} = c_t] = \lim_{x \downarrow c_t} \mathbb{E}[Y_{it}|X_{it} = x] - \lim_{x \uparrow c_t} \mathbb{E}[Y_{it}|X_{it} = x] \quad (1)$$

The continuity assumption could be violated if workers or their employers were willing and able to manipulate earnings strategically and precisely to impact workers' treatment status. In this case, potential discontinuities in observable and unobservable characteristics at the cutoff could confound estimates of the causal effects of the mandate. I discuss and test the validity of the identifying assumption in Section 4.3.

Estimation. Following the standard advice of the methodological RD literature (see in particular [Gelman and Imbens, 2019](#)), I estimate the target parameter β_{RD} by $\hat{\beta}_1$ from a local linear regression, allowing the slope of the control functions to vary on each side of the cutoff. The main specification is given by

$$Y_{it} = \beta_0 + \beta_1 \times \mathbb{1}\{X_{it} \geq c_t\} + \beta_2(X_{it} - c_t) + \beta_3(X_{it} - c_t) \times \mathbb{1}\{X_{it} \geq c_t\} + Z'_{it}\gamma + \epsilon_{it}, \quad (2)$$

where Y_{it} is a savings outcome of interest for individual i in year t , X_{it} denotes earnings in the main job (the running variable), c_t is the mandate cutoff determining treatment assignment, and ϵ_{it} is the error term. Z_{it} represents a vector of controls consisting of year fixed effects, age fixed effects, gender, and marital status. Control variables are not necessary for identification, but their inclusion improves precision (Calonico et al., 2019). The treatment variable is $\mathbb{1}\{X_{it} \geq c_t\}$ which indicates being subject to the pension plan mandate. Provided that ϵ_{it} does not change discontinuously at the threshold, $\hat{\beta}_1$ provides an unbiased estimate of the average treatment effect of the mandate on savings outcome Y_{it} for workers in proximity of the cutoff.

To consistently decompose the effect on total savings across the portfolio, the main analysis uses a bandwidth of CHF 10,000 for all outcomes.¹⁷ To give more weight to observations close to the cutoff, I use a triangular kernel. To account for potential sorting around the threshold, the main analysis employs a “donut hole” approach excluding workers whose earnings fall within CHF 700 of the threshold (Bajari et al., 2011; Barreca et al., 2011; Barr, Eggleston and Smith, 2022). This amount corresponds to the maximum mandatory pension contribution for workers just above the threshold.¹⁸ The idea is that potential sorting should be local: even if workers cared only about disposable income, they would still prefer gross earnings above the threshold, despite being subject to the mandate, over gross earnings more than CHF 700 below the cutoff. Hence, for workers remaining in the sample, manipulation of earnings to affect treatment status is very unlikely. This restriction also resolves the issue that gross earnings of workers within CHF 350 of the threshold cannot be imputed unambiguously.¹⁹

Inference. I compute standard errors using the heteroskedasticity-robust nearest-neighbor variance estimators proposed by Calonico, Cattaneo and Titiunik (2014a) and implemented in statistical software by Calonico, Cattaneo and Titiunik (2014b) and Calonico et al. (2017), as they are typically more robust in finite samples than conventional Huber-Eicker-White heteroskedasticity-robust standard errors (Calonico, Cattaneo and Titiunik, 2014a).

Sample Restrictions. The main analysis uses data from 2005 to 2017. I exclude 2002–2004 because the 2005 reform changed a number of provisions related to the

¹⁷I have also implemented the data-driven bandwidth selectors proposed by Calonico, Cattaneo and Titiunik (2014a,b). However, their procedure suggests very small bandwidths that, in combination with the donut approach, leave few observations for estimation and result in large standard errors. Instead, I use a consistent bandwidth across all outcomes and verify the robustness of the results with respect to the bandwidth choice in Section 5.5.

¹⁸During the sample period, the maximum mandatory pension contributions for workers just above the threshold are: $18\% \times \text{CHF } 3,525 = \text{CHF } 634.50$

¹⁹Estimation is performed using the Stata package `rdrobust` developed by Calonico, Cattaneo and Titiunik (2014b); Calonico et al. (2017).

mandate, including substantially lowering the threshold and changing the contribution rates for women (see Appendix Section B.4 for more information on the reform). As the pension plan mandate only applies to workers from the age of 25, I restrict the estimation sample to those aged 25–60.²⁰ Further, I exclude anyone receiving more than CHF 25,000 in self-employment income because individuals whose main source of income is self-employment can request to be exempt from the mandate even if they have some earnings from employment. Finally, I remove individuals who are not in the data in the preceding year because private savings cannot be computed for them (4% of the sample, most likely people moving to the canton from elsewhere). This leaves me with an effective estimation sample of 597,120 observations from 191,400 unique individuals.

Appendix Table A.1 presents summary statistics for the analysis sample. The average worker is 43 years old. 82% of workers in the sample are women, 67% are married. The average worker receives CHF 25,300 in total income, most of which represents earnings from their main job. Mean total savings are CHF 4,600 per year. Private savings account for 69%, whereas voluntary pension savings account for 26% of total savings. Mandatory pension savings make up 5% of total savings, on average, but more than 10% among workers above the mandate threshold.

Robustness Checks. In Section 5.5, I demonstrate that the estimation results are robust to making different operationalization choices regarding the bandwidth, size of the donut hole, kernel function, control variables, winsorization, and years of data included in the estimation sample.

4.3 Validity of Identifying Assumptions

The key identifying assumption is that there is a discontinuity in mandatory pension contributions at the threshold, whereas average potential outcomes are continuous. The main threat to identification stems from employees or employers manipulating earnings to sort around the threshold. The Swiss government usually announces the mandate threshold between the middle of September and the middle of October before it comes into force on the 1st of January. To affect their treatment status, employees would need to adjust their working hours, negotiate a change in the wage rate, or switch into a new job. Taking on a second job would not change treatment status because the mandate threshold applies to earnings in the main job only. Employers might try to push the earnings of their employees below the threshold in order

²⁰I impose a lower maximum age threshold than the legal retirement age – 64 years for women, 65 years for men – to keep the age range consistent across genders and to exclude people who retire early. Early retirement is widespread in Switzerland (Dorn and Sousa-Poza, 2005). I use the age group 20–24 years in a placebo check described in Section 4.3.

to avoid paying the employer share of mandatory pension contributions (assuming some of the incidence falls on them).

To rule out sorting around the threshold, the main analysis excludes workers who have earnings within CHF 700 of the threshold. As this exceeds the maximum mandatory pension contribution of workers just above the threshold, endogenous sorting is very unlikely to occur outside that range. Nevertheless, I assess the plausibility of the identifying assumptions by examining the continuity of the density of the running variable and predetermined covariates across the threshold, and by estimating saving responses for a placebo sample of workers aged 20–24 who are too young to be subject to the mandate.

Density of Running Variable. Following [McCrary \(2008\)](#), I examine the continuity of the density of the running variable across the threshold. Appendix Figure [A.2](#) shows the frequency distribution of annual earnings in the main job within the estimation window. Workers with earnings within CHF 700 of the threshold, who are excluded from the main analysis, are represented by the light blue bars. As noted before, I remove workers with gross earnings within CHF 350 of the threshold because their gross earnings cannot be imputed unambiguously.

The density looks relatively smooth and has a similar level below and above the mandate threshold. There is not much indication of bunching around the cutoff. While the number of workers just below the threshold is slightly higher than just above, the difference is minor relative to the overall frequencies, particularly when abstracting from the CHF 700 range around the threshold which is omitted from the main estimation sample. This is in line with previous findings that employees fail to precisely sort around earnings thresholds to take advantage of financial incentives ([Chetty et al., 2011](#); [Chetty, Friedman and Saez, 2013](#); [Chetty et al., 2014](#)).

Continuity of Predetermined Characteristics. As endogenous sorting around the cutoff would likely result in imbalances on covariates, I assess the continuity of a set of predetermined characteristics across the threshold. I implement this balance test by estimating the main RD specification in Equation (2) using age, gender, marital status, number of children, or household income as the dependent variable (controlling only for year fixed effects).

Appendix Figure [A.3](#) displays the results. I find that workers with earnings marginally below and above the threshold have very similar characteristics. Based on the precision of the estimates, I can rule out even small discontinuities for gender, marital status, number of children, and household income. I do find a significant discontinuity in age, implying that workers just above the threshold are on average a quarter of a year younger than those narrowly below. While the magnitude of this difference is modest, I address this potential source of bias by controlling for age

fixed effects in the main analysis.

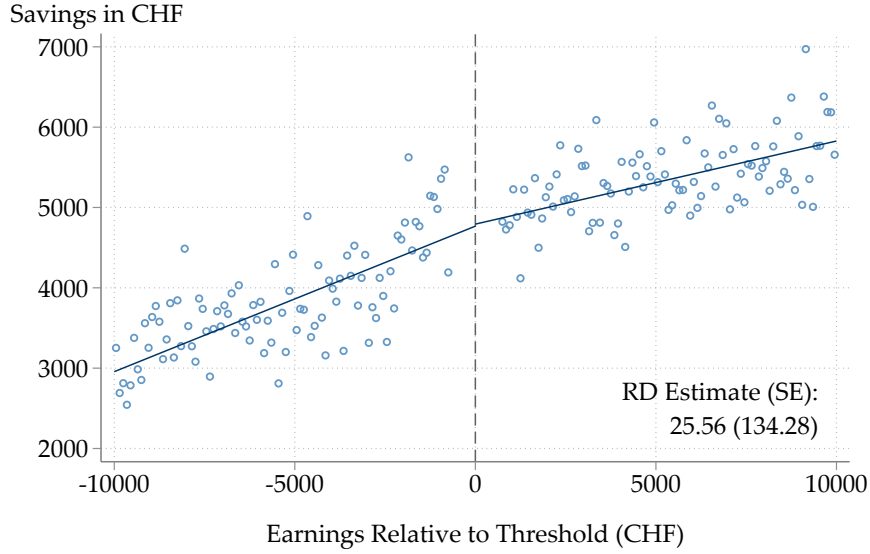
Placebo Sample. Based on the smoothness of the density and covariates across the threshold, endogenous sorting does not appear to be a major concern. But continuity of potential outcomes can also be violated if there are other discontinuities at the threshold that affect savings, e.g., if the same cutoff value is used in another policy (Imbens and Lemieux, 2008). Ex ante, this seems highly unlikely as I am not aware of any other policies targeting the same cutoff and it moves from year to year between specific non-round numbers, defined as the part of the salary that is covered by the pay-as-you-go old-age insurance system (see Appendix B for more details).

I can test this empirically, taking advantage of the fact that the mandate only affects workers from the year in which they turn 25. In a placebo exercise, I restrict the sample to workers aged 20–24 and re-run the main regression discontinuity estimation. As these workers are not subject to the mandate, I should not find a saving response in the data. Appendix Figure A.4 shows the estimated placebo effects on total savings, private savings, and voluntary pension savings. In contrast to the main worker sample, I estimate precise null effects on all three savings outcomes among this group of young workers unaffected by the mandate. This is evidence that there is no other discontinuity at the threshold other than the mandate, at least none that affects both young workers and older workers.

5 Saving Responses to Mandatory Pension Plans

As described above, the pension plan mandate forces the average worker narrowly above the threshold to contribute around CHF 400 per year – roughly 2% of their earnings – to an employer-sponsored pension account. This section presents the results from the regression discontinuity analysis leveraging this variation in mandatory pension savings. I begin by presenting estimates of the mandate’s effect on total savings. Then, I investigate potential portfolio reallocation responses, estimating separate effects on private savings and its components, including savings in financial assets, business assets, property, other wealth, and debt, as well as voluntary pension savings. This enables me to identify the specific asset types that drive the response. Putting all the estimates together, I characterize the saving response to mandatory pension plans across the entire portfolio. Finally, I conduct a range of robustness checks to probe the sensitivity of the results.

FIGURE 2: EFFECT OF PENSION PLAN MANDATE ON TOTAL SAVINGS



Notes: Regression discontinuity plot showing the effect of the pension plan mandate on total savings. Effective estimation sample includes 597,120 worker-year observations from 2005 to 2017. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel. The running variable is recentered around the mandate threshold, indicated by the dashed vertical line. Total savings are the sum of mandatory pension savings, private savings, and voluntary pension savings. Private savings are winsorized at percentiles 5 and 95. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

5.1 Total Savings

Figure 2 plots total savings against gross earnings in the main job (i.e., the running variable), providing graphical evidence of the mandate's effect on overall savings. The dots are local sample means of 100 non-overlapping, evenly spaced bins on each side of the cutoff. The lines are linear fits from a non-parametric regression using all underlying datapoints and a triangular kernel. The point estimate and standard error for the treatment effect are displayed in the figure, obtained from estimating the RD specification in Equation (2).

Strikingly, there is no discontinuity at the cutoff, indicating that the mandate has little impact on the overall savings level of workers. The point estimate of CHF 26 is not statistically significantly different from zero and corresponds to a crowd-out rate of 94%. Based on the 95% confidence interval, I can rule out increases in total savings exceeding CHF 290, or equivalently, crowd-out rates lower than 29%. The headline estimate implies that mandatory pension contributions are offset by reductions in other types of savings, providing evidence of strong portfolio reallocation responses to the mandate. I proceed by decomposing the overall effect to reveal the type of

savings that workers adjust to undo the mandate’s impact.

5.2 Private Savings

Figure 3 displays the effects of mandatory pension plans on private savings, which include all forms of non-pension savings, and its components.

Overall Private Savings. Panel A plots the response of overall private savings. I find that private savings drop by CHF 459 at the cutoff, which is statistically highly significant. The magnitude of this effect is close to the average mandatory retirement contributions of roughly CHF 400 among workers above the threshold. This implies that workers respond to the mandate by lowering private savings which offsets its effect on total savings.

Private Savings by Asset Type. Drawing on detailed wealth microdata from the tax records, I investigate what types of assets account for the large decline in private savings. Panels B through F present the effect of the mandate on savings in financial assets, business assets, property, other types of wealth, and debt, respectively.²¹ The results indicate that the overall drop is mainly driven by reduced savings in financial assets. As Panel B shows, the mandate lowers financial savings by CHF 311, again statistically highly significant. Financial assets include bank accounts, stocks, bonds, and other securities. While I cannot disaggregate financial assets further to probe whether the drop in savings is driven by bank accounts or more long-term investments, it seems plausible that workers use more liquid forms of wealth to fund mandatory pension contributions.

Panels C through F show that the mandate does not affect savings in business wealth, property wealth, other types of wealth, or debt. Three out of four point estimates are negative but they are statistically indistinguishable from zero. For savings in property, I find a decrease by CHF 80 but the confidence interval is relatively wide because property investments are lumpy. The effect estimates for business wealth, other wealth, and debt are an order of magnitude smaller than the effect on financial savings, and they are sufficiently precise to reject even small changes. The lack of an effect on debt is in line with [Beshears et al. \(2022\)](#) who do not find an impact of automatic enrollment on debt or credit scores in the US, but differs from [Beshears et al. \(2024\)](#) and [Choukhmane and Palmer \(2025\)](#) which both find that workers in the UK partially finance higher pension savings induced by automatic enrollment through an increase in debt.

²¹“Other Wealth” includes an individual’s share of assets in joint heirship or other types of co-ownership as well as other assets such as cash, gold, art, boats, etc.

FIGURE 3: EFFECTS OF PENSION PLAN MANDATE ON PRIVATE SAVINGS



Notes: Regression discontinuity plots showing the mandate's effect on private savings. Effective estimation sample includes 597,120 worker-year observations from 2005 to 2017. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel. All outcomes are winsorized at percentiles 5 and 95, except savings in business wealth and property wealth which are winsorized at percentiles 2.5 and 97.5.

Source: Author's calculations based on administrative tax data from the canton of Bern.

5.3 Voluntary Pension Savings

Mandatory pension plans could also affect voluntary retirement savings, which mainly consist of private pension contributions to third-pillar accounts.²² Because they are deductible from taxable income, I can directly observe voluntary pension savings in the tax data. Below, I first estimate the overall effect of the pension plan mandate on voluntary pension savings. As both mandatory pension savings and the cap on preferentially taxed private pension savings increase discontinuously at the mandate threshold, the overall effect reflects a combination of their potential impacts.²³ Subsequently, I disentangle the effect of mandatory pension contributions from the impact of the cap. This enables me to assess the degree of substitution between mandatory and voluntary pension savings.

Overall Voluntary Pension Savings. Figure 4 shows that workers respond to the mandate by *increasing* their voluntary retirement savings by CHF 112. The lower bound of the 95% confidence interval is CHF 79. While voluntary pension savings generally increase linearly with earnings, with similar slopes below and above the cutoff, there is a clear jump upwards at the threshold. The direction of the effect is surprising at first sight, as substitution between mandatory and voluntary pension savings would imply a negative sign, but it is contaminated by the increase in the cap on private pension contributions at the threshold.²⁴ Next, I isolate the impact of mandatory pension savings.

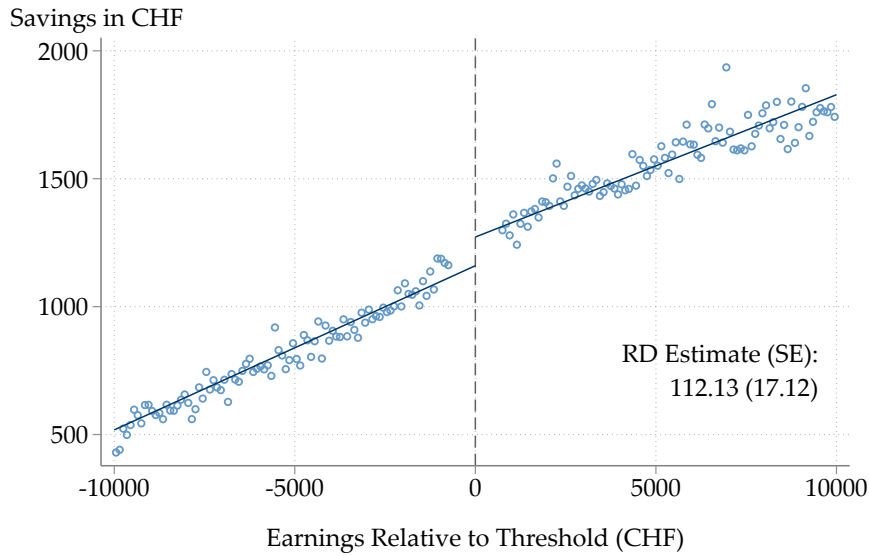
Isolating the Impact of Mandatory Pension Savings. I disentangle the effect of mandatory pension savings from the impact of the contribution cap in two complementary exercises: First, I estimate the effects of the mandate along the distribution of private pension savings, exploiting that the two forces should have distinct impacts on different parts of the distribution: while mandatory pension savings should, if anything, decrease private pension savings across the entire distribution, including at the extensive margin, the higher contribution cap should raise private pension savings, but only at the upper end of the distribution where the cap is binding. In the second exercise, I isolate the effect of mandatory pension savings by imposing the same contribution cap for workers above and below the threshold and re-estimating the effect on private pension savings.

²²Private pension savings account for 94% of voluntary pension savings among workers in the estimation sample (see Appendix Table A.1). Voluntary buy-ins into occupational pension plans are of minor importance, as only 0.7% of worker-year observations make non-zero buy-ins.

²³As explained in more detail in Section 2.2, workers just below the threshold face an annual cap on private pension savings of CHF 3,000–4,000, whereas those just above the threshold can contribute up to CHF 6,000–7,000, with the exact amount depending on the year.

²⁴Appendix Figure A.5 shows that private pension savings indeed account for the entire effect on overall voluntary pension savings.

FIGURE 4: EFFECT OF PENSION PLAN MANDATE ON VOLUNTARY PENSION SAVINGS



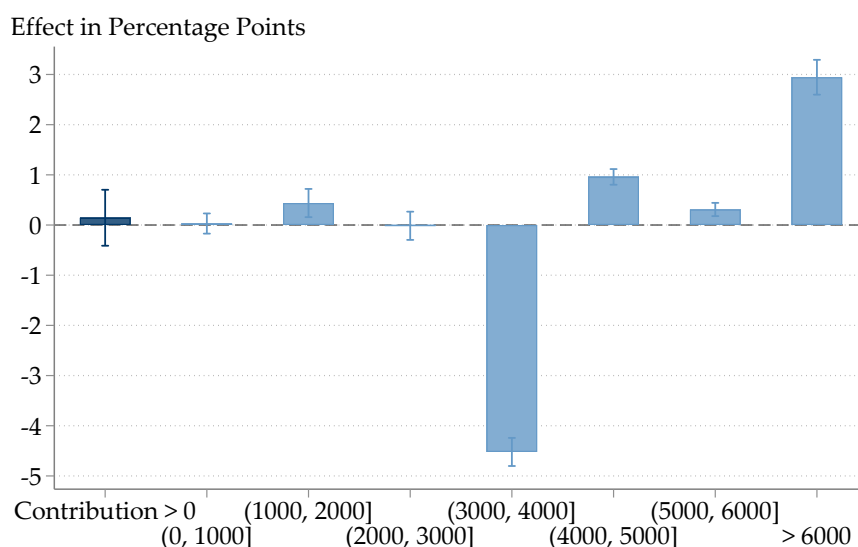
Notes: Regression discontinuity plot showing the effect of the pension plan mandate on voluntary pension savings. Effective estimation sample includes 597,120 worker-year observations from 2005 to 2017. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel. The running variable is recentered around the mandate threshold, indicated by the dashed vertical line. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

Figure 5 presents the results of the first exercise, plotting estimated RD effects on the share of workers making non-zero private pension contributions (extensive margin) in dark blue and the share contributing amounts in intervals of CHF 1,000 in light blue.²⁵ They suggest that the effect of the mandate on voluntary pension savings is entirely driven by the increase in the contribution cap at the threshold. At the extensive margin and at the lower end of the distribution, where the cap is irrelevant, effect estimates are close to zero. This implies that there is little substitution between mandatory and voluntary pension savings. By contrast, the share of workers with private pension savings in the interval where the cap below the threshold is located (CHF 3,000–4,000) declines by 4.5 percentage points. Conversely, the share contributing more than CHF 4,000 increases by almost 4.5 percentage points. This can be interpreted as the share of workers near the threshold for whom the lower contribution cap is binding. The share contributing CHF 4,000–5,000 increases by 1 percentage point, the share contributing CHF 5,000–6,000 by 0.3 percentage points,

²⁵Appendix Figure A.6 displays the corresponding RD plots. They clearly document the bunching of workers below the threshold at the lower contribution cap. For example, Panel E shows that a sizeable share of workers with earnings just below the cutoff (i.e., with gross earnings of around CHF 20,000) make contributions of CHF 3,000–4,000 which is where their cap is located. This share is much lower above the cutoff where the cap is higher.

FIGURE 5: EFFECT OF MANDATE ON DISTRIBUTION OF PRIVATE PENSION CONTRIBUTIONS



Notes: Coefficient plot showing the effect of the pension plan mandate on the share of workers making private pension contributions of different magnitudes indicated on the x-axis. Effective estimation sample includes 597,120 worker-year observations from 2005 to 2017. Point estimates and 95% confidence intervals are obtained from estimating Equation (2) using a triangular kernel. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

and the share contributing more than CHF 6,000 by 3 percentage points.

The results of the second exercise confirm that the mandate does not affect voluntary pension savings once the contribution cap is held fixed. As Appendix Figure A.7 shows, I estimate a precise null effect when limiting private pension contributions of all workers to at most 20% of their net earnings and estimating the effect of the mandate on this capped outcome. This supports the conclusion that there is no substitution between mandatory and voluntary pension savings.

Occupational Pension Buy-ins. Panel B of Appendix Figure A.5 depicts the effect of the mandate on voluntary lump-sum buy-ins into occupational pension plans, which are not subject to the cap on private pension contributions. I also find a precisely estimated null effect for this type of voluntary pension savings, allowing me to rule out changes by more than CHF 30 at the 5% level. Again, this is consistent with a lack of substitution between mandatory and voluntary pension savings.

5.4 Summary of Saving Responses Across the Portfolio

Mandatory pension plans have limited effects on total savings, on average, as workers offset mandatory contributions by reducing private savings in financial assets. Savings in other non-pension assets are not significantly affected. Voluntary pension

savings also remain unchanged once the contribution cap is held fixed. Overall, the mandate induces a reallocation of savings from more liquid financial assets to less liquid retirement accounts.

In the theoretical framework of [Chetty et al. \(2014\)](#), an increase in overall retirement savings with no change in total savings is consistent with a large share of workers behaving like passive savers with a consumption target. Viewed through the lens of the model, these workers do not re-optimize their voluntary pension savings in response to higher mandatory contributions. Instead, they keep consumption spending fixed, so the resulting drop in disposable income leads to substantial crowding-out.

5.5 Robustness Checks

In this section, I check the sensitivity of the estimated effects on the main outcomes – total savings, overall private savings, and overall voluntary pension savings – to different choices regarding the RD specification in Equation (2) and its operationalization. I show robustness with respect to changing the bandwidth, size of the donut hole, kernel weights, set of control variables, winsorization cutoffs, and the years of data included in the estimation sample.

Bandwidth. Appendix Figure [A.8](#) plots point estimates and 95% confidence intervals for the effects of the mandate, varying the bandwidth symmetrically around the main bandwidth of CHF 10,000 – from CHF 2,000 to CHF 18,000. Overall, the findings are robust to using alternative bandwidths. Panel A documents that all bandwidths yield estimated effects on total savings that are close to zero and statistically insignificant, except for the CHF 4,000 bandwidth which produces a negative effect estimate. Thus, there is no bandwidth at which the mandate is found to significantly increase total savings. As Panel B shows, I estimate a significant negative effect on private savings for all bandwidths except the narrowest, which yields a point estimate comparable to the main result but with a substantially wider confidence interval that includes zero. Panel C shows that the positive effect on voluntary pension savings remains relatively stable for bandwidths of CHF 6,000 or more. For smaller bandwidths, the point estimates decline and the standard errors increase, making the estimates statistically indistinguishable from zero.

Size of Donut Hole. To rule out sorting around the threshold, the main analysis excludes workers within CHF 700 of the cutoff, as sorting is implausible to occur outside that range (see Section 4.2). Appendix Figure [A.9](#) examines the sensitivity of the estimates to the width of the donut hole that is removed from the estimation sample. When varying the donut hole from zero to CHF 1,400, the estimates remain consistent with the main results. The estimated effects on total savings (Panel A)

and private savings (Panel B) are very stable for donut sizes up to CHF 1,000. They increase modestly for larger donut holes, but all confidence intervals continue to include the main point estimate. Panel C shows that the estimated effects on voluntary pension savings are always positive and highly statistically significant.

Kernel. Appendix Figure A.10 shows that the results are robust to using alternative weighting functions in the local linear regression. For all outcomes, using a uniform or Epanechnikov kernel yields point estimates and confidence intervals that are very similar to those obtained with a triangular kernel.

Control Variables. Appendix Figure A.11 displays the saving responses estimated using different sets of controls. Including only year fixed effects, or omitting controls entirely, yields estimates that are very similar to those from the main specification which controls for year fixed effects, age fixed effects, marital status, and gender.

Winsorization. Following García-Miralles and Leganza (2024), I winsorize private savings at percentiles 5 and 95. To verify that the findings are not sensitive to the winsorization cutoffs, Appendix Figure A.12 compares the main results to estimates obtained when winsorizing private savings at percentiles 1 and 99 or percentiles 10 and 90 (as in Chetty et al., 2014) instead. Although the width of the confidence intervals naturally varies with the degree of winsorization, the different estimates consistently point to sizeable reductions in private savings which limit the effect of mandatory pension contributions on total savings.

Sample Period. The main analysis sample excludes years 2003 and 2004 because a reform implemented in 2005 lowered the mandate threshold and changed the contribution schedule for women.²⁶ As shown in Appendix Figure A.13, including these years in the estimation sample yields very similar results. The estimates for total savings and private savings are virtually unchanged. The positive effect on voluntary pension savings is slightly attenuated but remains sizeable and highly significant. The attenuation is expected, as the discontinuity in the contribution cap at the threshold, which drives the entire effect, was smaller prior to the reform.²⁷

6 Saving Responses Under Liquidity Constraints

The average worker responds to mandatory pension plans by sharply cutting private savings, especially in financial assets. But workers with limited liquidity may not

²⁶Note that the tax data extend back to 2002, but I exclude that year from the analysis because private savings cannot be calculated without observing wealth in the previous year.

²⁷As the 2005 reform lowered the threshold by around CHF 6,000, this reduced the contribution cap just below the cutoff (20% of net earnings) by more than CHF 1,000.

be able to reduce their private savings to offset the mandate's impact. In this section, I assess the savings effects of the mandate on workers who are subject to such liquidity constraints. I explore the role of liquidity by allowing for heterogeneous responses across workers with different amounts of financial wealth or household income.²⁸ The idea is that drawing down liquid assets or resource sharing within the household can enable workers to hold consumption constant despite larger pension contributions. To implement this, I split up the estimation sample by terciles of financial wealth in the previous year (as current wealth may be affected by treatment status) or household income, respectively, and estimate the effect of the mandate on total savings, private savings, and voluntary pension savings separately in these subsamples.²⁹ Reassuringly, I obtain similar results when using financial assets or household income as a proxy for liquidity constraints.

To the extent that saving decisions are made jointly within the household, heterogeneity by household income also sheds light on how saving responses vary across the income distribution. This allows me to investigate whether the regression discontinuity estimates for workers with earnings near the mandate threshold have external validity for higher-income individuals.

Total Savings. Panel A of Figure 6 plots the estimated effects on total savings for subsamples of workers who differ in their likelihood to be affected by liquidity constraints. While splitting the sample reduces precision, the pattern of point estimates is consistent with the notion that liquidity-constrained workers cannot fully offset mandatory pension contributions by reducing other savings. For workers in the bottom tercile, I estimate that the mandate increases total savings by CHF 170 when using financial wealth and by CHF 350 when using household income as a proxy. Contrary to the limited average effect, this provides evidence that the mandate increases total savings of workers who are not able to finance forced pension contributions by reducing liquid assets or receiving financial support from their partner. By contrast, the point estimates for total savings are close to zero for workers in the top tercile, implying full crowding-out in the absence of liquidity constraints.

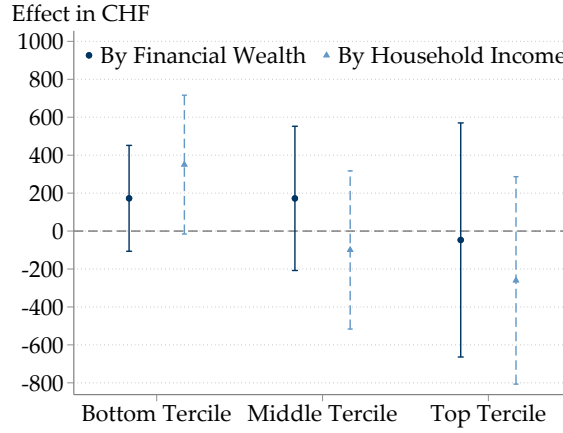
Private Savings. Panel B presents the estimated effects on private savings. As expected, the larger effect on total savings in the bottom tercile is driven by a weaker negative response in private savings among these plausibly liquidity-constrained

²⁸Household income includes all types of income that an individual or married couple receive in a given year. Besides their own earnings, the spouse's earnings from employment are by far the most important income source for workers in the estimation sample. 67% of them are married, with partners receiving CHF 79,500 in employment income, on average. Averaging across all workers, with and without a partner, mean household income is CHF 79,800, of which 66% is accounted for by the partner's employment income and 27% by the worker's own employment income.

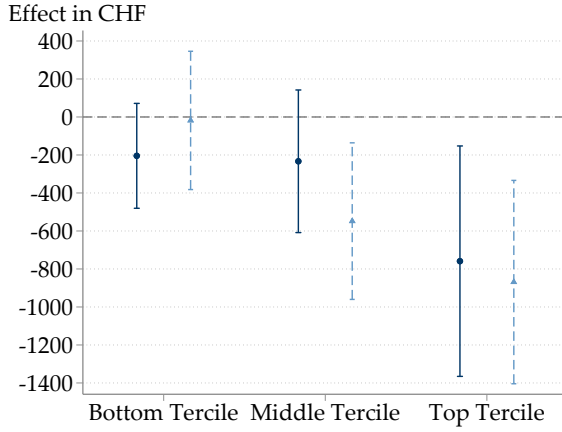
²⁹The tercile cutoff values for financial wealth are CHF 4,700 and CHF 33,000. The tercile cutoffs for household income are CHF 46,100 and CHF 100,200.

FIGURE 6: HETEROGENEITY IN SAVING RESPONSES BY LIQUIDITY CONSTRAINTS

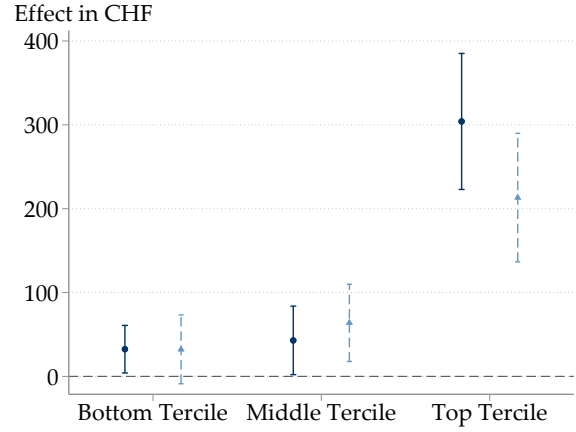
(A) TOTAL SAVINGS



(B) PRIVATE SAVINGS



(C) VOLUNTARY PENSION SAVINGS



Notes: Coefficient plots showing the effects of the pension plan mandate by tercile of financial wealth in the previous year and household income, respectively. Overall sample includes 597,120 worker-year observations from 2005 to 2017. Point estimates and 95% confidence intervals are obtained from estimating Equation (2) using a triangular kernel, separately for the tercile groups. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

workers. The estimated effect is around CHF –200 when using financial wealth and CHF –20 when using household income as a proxy. Neither estimate is statistically distinguishable from zero. For workers in the top tercile, I estimate a large and statistically significant decline in private savings by around CHF 800. Strikingly, the magnitude of this reduction exceeds the mandatory pension contributions.

Voluntary Pension Savings. Panel C displays heterogeneous effects on voluntary pension savings. The increase in voluntary pension savings in response to the mandate is highly concentrated among workers in the top tercile by financial wealth or

household income. This explains the disproportionate decline in private savings in the top tercile: these workers are not only offsetting mandatory pension contributions but also reallocating savings into preferentially taxed private pension accounts to take advantage of the higher contribution cap. For workers lower in the distribution of financial wealth or household income, I estimate much smaller increases in voluntary pension savings. Presumably, this is due both to liquidity constraints and to lower marginal income and wealth tax rates, which reduce the incentive to make these contributions.

Summary and Discussion. I find evidence that liquidity-constrained workers are less able to reduce private savings to offset the pension contributions required by the mandate. As a consequence, the mandate increases their total savings by around CHF 170–350, corresponding to roughly 1–2% of earnings. Workers with high levels of financial wealth or household income respond to the mandate (and the associated increase in the private pension contribution cap) by disproportionately decreasing private savings to offset the mandated pension contributions and reallocate savings into tax-advantaged private pension accounts, while holding overall savings constant. These results are consistent with [Choukhmane and Palmer \(2025\)](#) who find that workers with low initial deposit balances respond to increased default pension contributions mainly by cutting spending, whereas workers with high balances shift savings into pension accounts without reducing spending much.

The finding that the estimated effect on total savings for workers with high household income is close to zero tentatively suggests that the strong crowding-out response generalizes to higher-income individuals. If anything, crowd-out rates appear to increase along the income distribution. Heterogeneous responses driven by liquidity constraints may help explain why previous research on mandatory pension plans has found limited crowding-out among Danish workers with very low annual earnings of around \$5,000 ([Chetty et al., 2014](#)) and stronger crowding-out among faculty at US universities ([Card and Ransom, 2011](#); [Friedberg, Leive and Cai, 2024](#)).

7 Conclusion

In half of all OECD countries, workers are required to contribute to a pension plan. But there is limited evidence whether these mandates increase overall savings, mainly due to a lack of comprehensive savings data. This paper presents novel findings on saving responses to mandatory pension plans across the entire portfolio, drawing on tax microdata on wealth and savings from the Swiss canton of Bern and leveraging a discontinuity in mandatory pension contributions for identification.

I find that the mandate has limited effects on total savings, as workers offset

mandatory pension contributions by strongly reducing private savings. The drop in private savings is primarily driven by financial assets. I do not find an impact on savings in other types of assets such as business wealth, property, other wealth, and debt. I estimate a precise null effect on voluntary pension savings once the change in the contribution cap on preferentially taxed private pension savings is accounted for. Consistent with standard models of saving under borrowing constraints, liquidity-constrained workers reduce their private savings by less, and hence moderately increase their total savings in response to the mandate. Workers with high levels of financial wealth or household income disproportionately decrease their private savings to offset the mandated pension contributions and reallocate savings into tax-advantaged private pension accounts.

The strong substitution between mandatory pension contributions and private *non-retirement* savings highlights that it is crucial to assess the effects of pension policies throughout the entire portfolio. Prior research that focuses on the impact of mandatory pension plans on retirement savings therefore estimates a lower bound for the overall crowd-out rate (Card and Ransom, 2011; Friedberg, Leive and Cai, 2024). The result that liquidity constraints weaken the crowding-out response may help explain why previous work studying workers at different points of the income distribution has found a broad range of savings effects (Card and Ransom, 2011; Chetty et al., 2014; Friedberg, Leive and Cai, 2024; Jensen et al., 2025). An important contextual feature of mandatory pension plans in Switzerland is that they are fully funded, with a strong and salient tax-benefit linkage. This likely contributes to their substitutability with private savings.

These findings have implications for pension policy. If mandatory pension plans are intended to raise overall savings levels, their effectiveness appears limited. The exception is liquidity-constrained workers, but their savings gains come at a particularly high cost in terms of marginal utility from current consumption. Despite the modest impact on total savings, the mandate may still improve financial preparedness for retirement by shifting savings from financial assets to less liquid pension accounts. However, it is unclear whether this portfolio reallocation makes workers better off, as the lower share of liquid assets leaves them more exposed to negative shocks during their working lives (Choukhmane and Palmer, 2025). In the long run, potential differences in after-tax rates of return between the two types of assets may also play an important role. A thorough examination of these welfare considerations is beyond the scope of this paper but constitutes a fruitful avenue for future research.

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Online Appendix

Saving Responses to Mandatory Pension Plans¹

David Burgherr²

October 2025

Contents

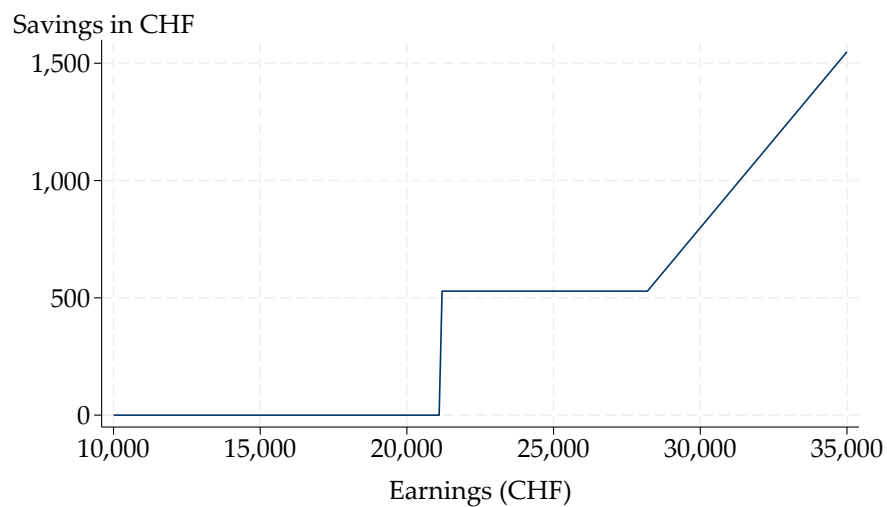
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A Additional Figures and Tables

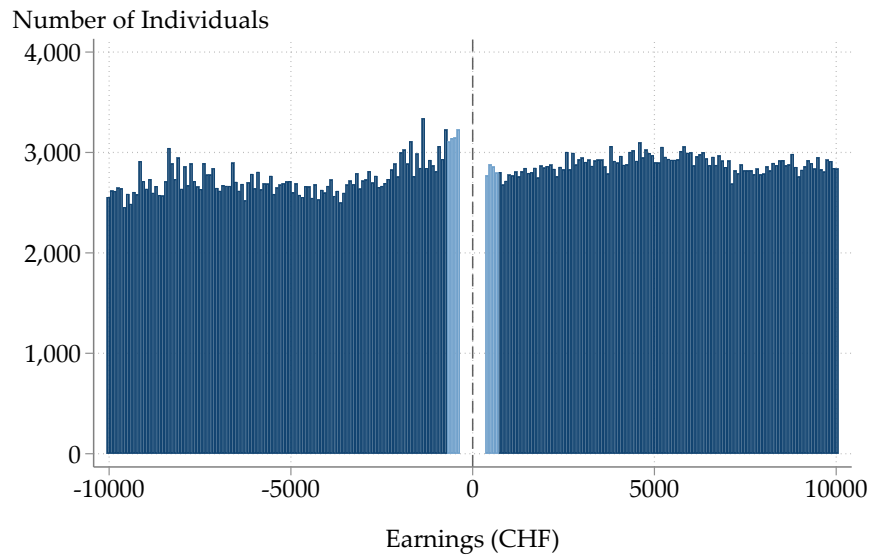
FIGURE A.1: MANDATORY PENSION CONTRIBUTIONS AS A FUNCTION OF EARNINGS



Notes: The figure displays mandated occupational pension contributions as a function of gross earnings for an employee between 45 and 54 years of age in 2017. Contributions are computed according to the statutory contribution schedule, i.e., applying a contribution rate of 15% to qualifying earnings. The relationship is very similar in other years, with slight variations depending on the year-specific parameters of the contribution schedule listed in Appendix Table B.1.

Source: Author's illustration based on information from the Federal Social Insurance Office.

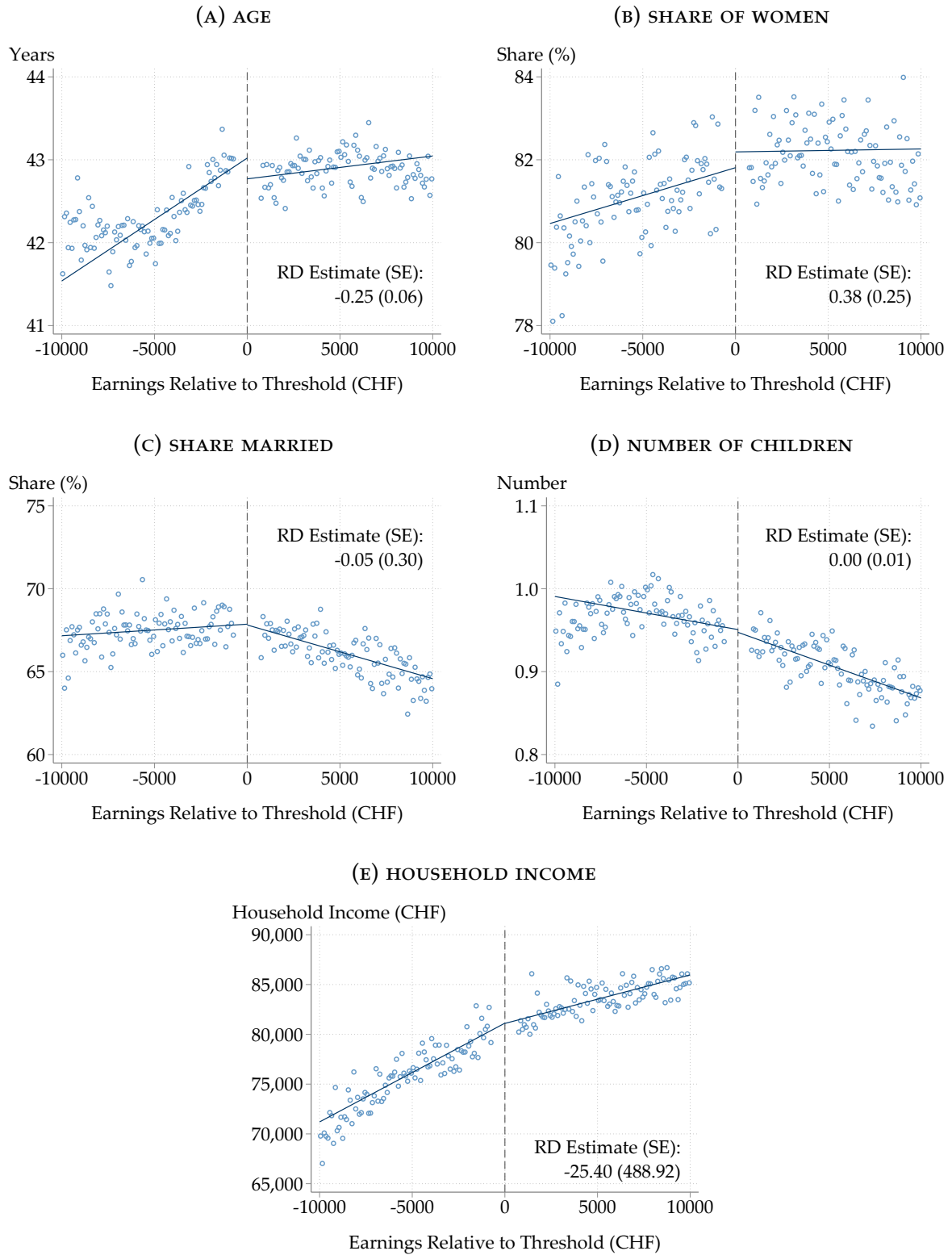
FIGURE A.2: FREQUENCY DISTRIBUTION OF EARNINGS IN MAIN JOB



Notes: Histogram representing the frequency distribution of gross earnings in the main job (running variable) within CHF 10,000 of the threshold of the pension plan mandate. The light blue bars represent workers with earnings within CHF 700 of the threshold who are removed from the main estimation sample; the dark blue bars represent workers included in the main estimation sample. The running variable is recentered around the mandate threshold, indicated by the dashed vertical line. The sample is based on data from 2005 to 2017. Workers within CHF 350 of the cutoff are excluded because their gross earnings cannot be imputed unambiguously. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions.

Source: Author's calculations based on administrative tax data from the canton of Bern.

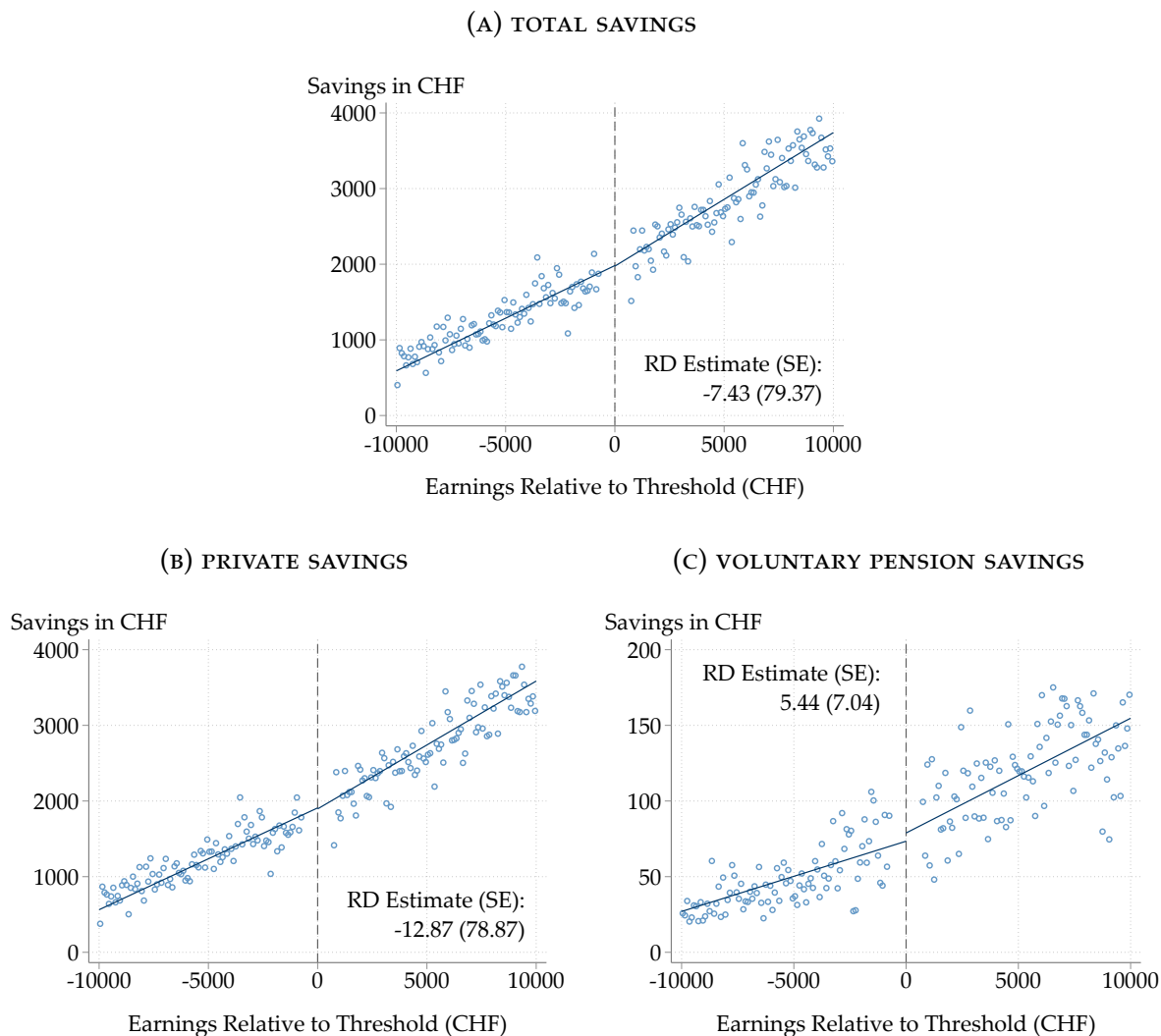
FIGURE A.3: CONTINUITY OF PREDETERMINED CHARACTERISTICS



Notes: Regression discontinuity plots for a set of predetermined characteristics. Effective estimation sample includes 597,120 worker-year observations from 2005 to 2017. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel and controlling for year fixed effects. The running variable is recentered around the mandate threshold. See Section 4.2 for more details on estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

FIGURE A.4: PLACEBO TEST: EFFECTS OF MANDATE ON WORKERS AGED 20–24

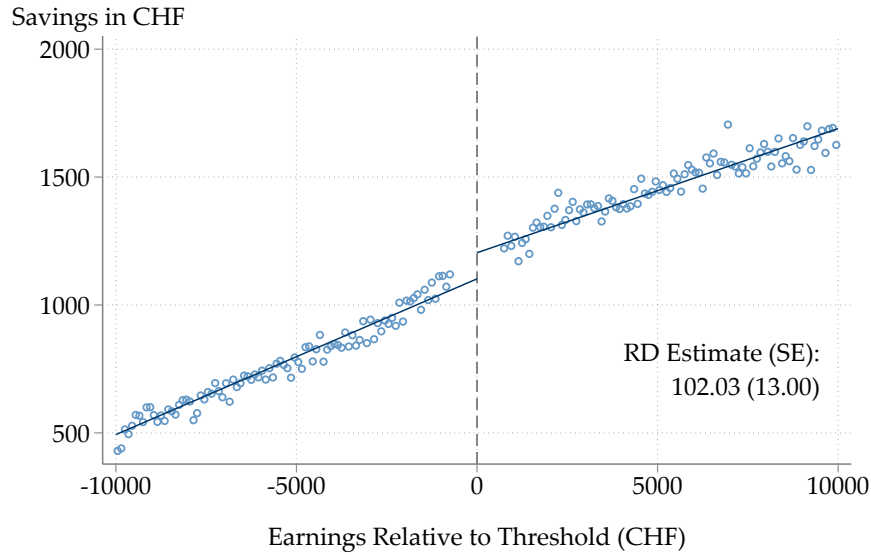


Notes: Regression discontinuity plots showing the effects of the pension plan mandate on a placebo sample of workers who are 20–24 years old and thus not affected by the mandate. Effective estimation sample includes 169,443 worker-year observations from 2005 to 2017. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel. The running variable is recentered around the mandate threshold, indicated by the dashed vertical line. Overall private savings are winsorized at percentiles 5 and 95. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on estimation.

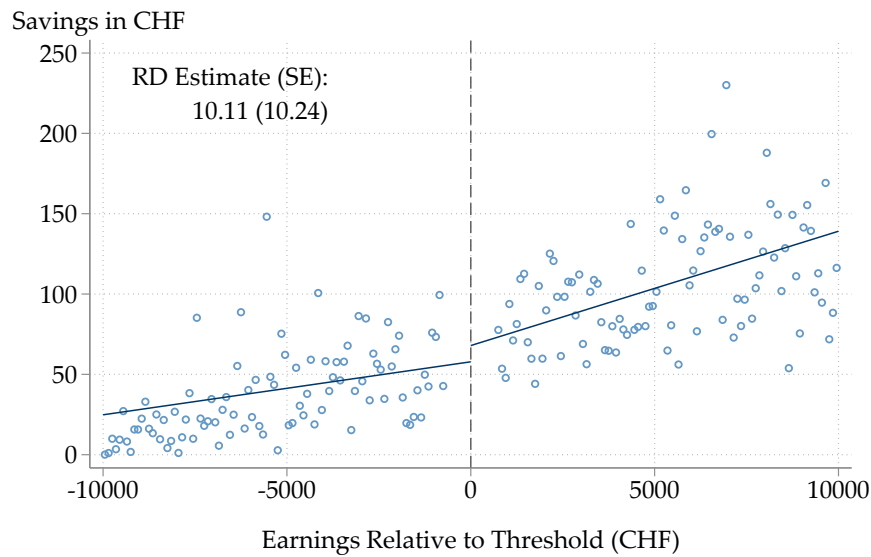
Source: Author's calculations based on administrative tax data from the canton of Bern.

FIGURE A.5: EFFECTS OF MANDATE ON COMPONENTS OF VOLUNTARY PENSION SAVINGS

(A) PRIVATE PENSION SAVINGS



(B) OCCUPATIONAL PENSION BUY-INS



Notes: Regression discontinuity plots showing the effect of the pension plan mandate on private pension savings (Panel A) and voluntary buy-ins into occupational pension plans (Panel B). Effective estimation sample includes 597,120 worker-year observations from 2005 to 2017. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel. The running variable is recentered around the mandate threshold, indicated by the dashed vertical line. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

FIGURE A.6: EFFECTS OF MANDATE ON INTERVALS OF PRIVATE PENSION CONTRIBUTIONS

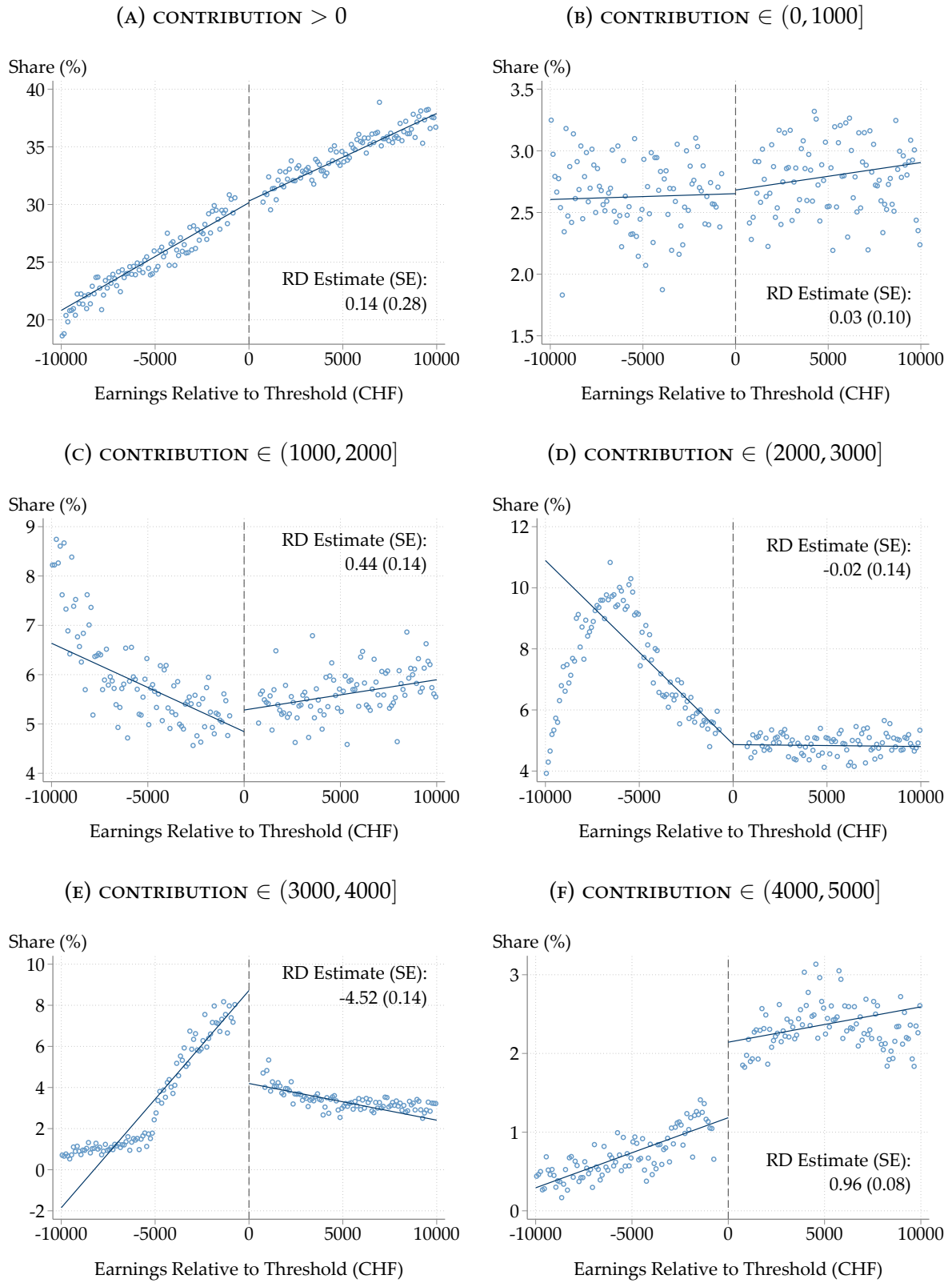
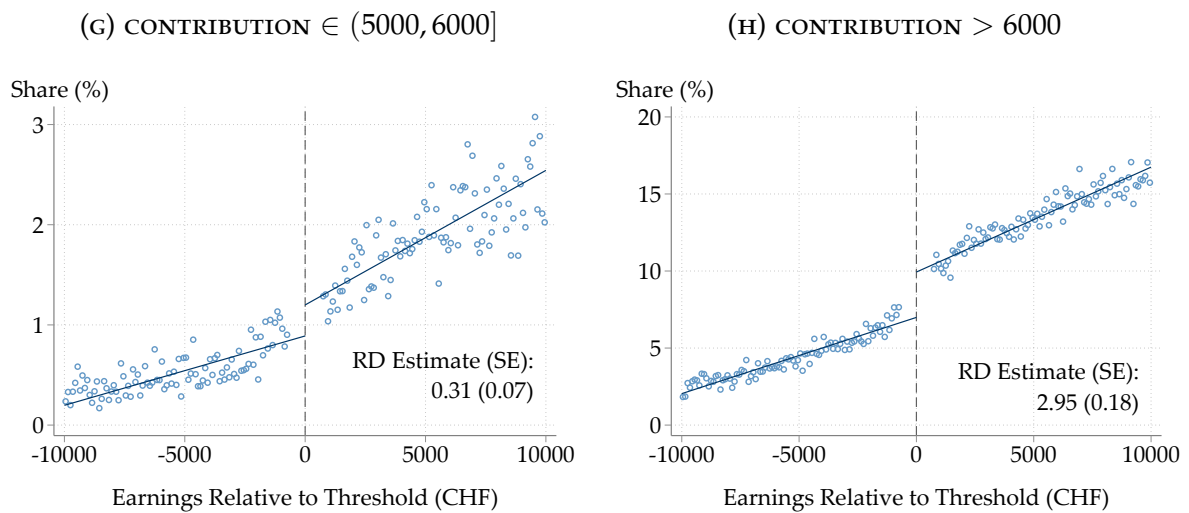


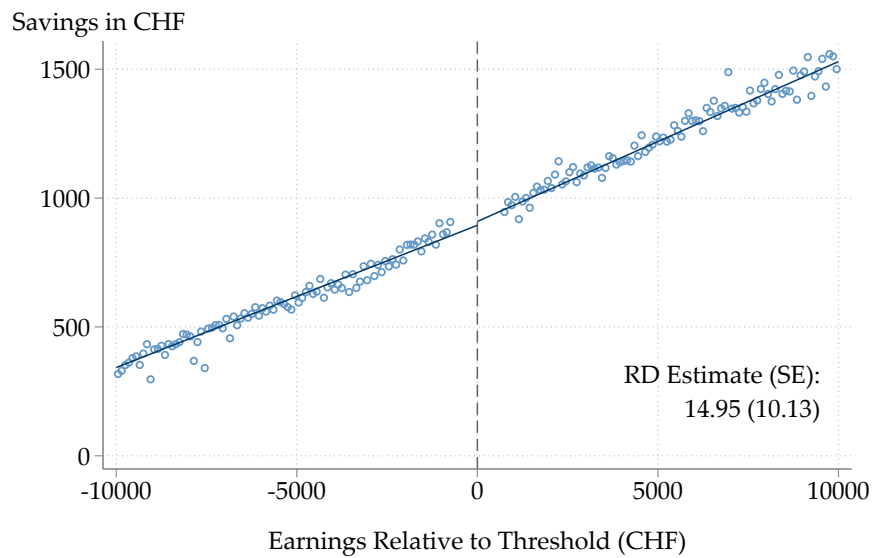
FIGURE A.6: EFFECTS OF MANDATE ON INTERVALS OF PRIVATE PENSION CONTRIBUTIONS (CONT.)



Notes: Regression discontinuity plots showing the effect of the pension plan mandate on the share of workers making any non-zero private pension contribution (Panel A) and on the share making contributions in CHF 1,000 intervals (Panels B through H). Effective estimation sample includes 597,120 worker-year observations from 2005 to 2017. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel. The running variable is recentered around the mandate threshold, indicated by the dashed vertical line. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

FIGURE A.7: EFFECT OF MANDATE ON CAPPED PRIVATE PENSION SAVINGS

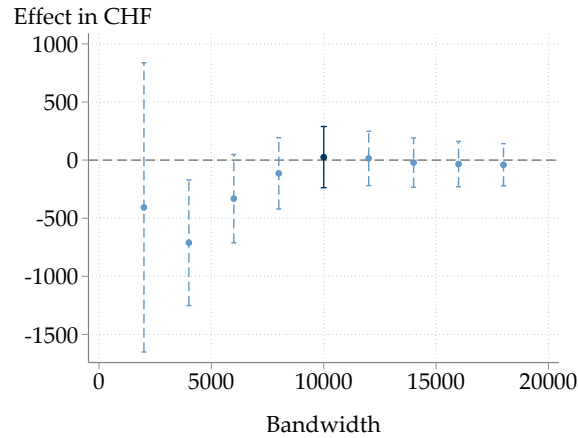


Notes: Regression discontinuity plot showing the effect of the pension plan mandate on private pension savings after imposing the same contribution cap rule below and above the mandate threshold. Effective estimation sample includes 597,120 worker-year observations from 2005 to 2017. Points are local sample means using 100 non-overlapping evenly-spaced bins on each side of the cutoff; lines are linear fits using all the underlying data points. Point estimates and standard errors are obtained from estimating Equation (2) using a triangular kernel. The running variable is recentered around the mandate threshold, indicated by the dashed vertical line. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

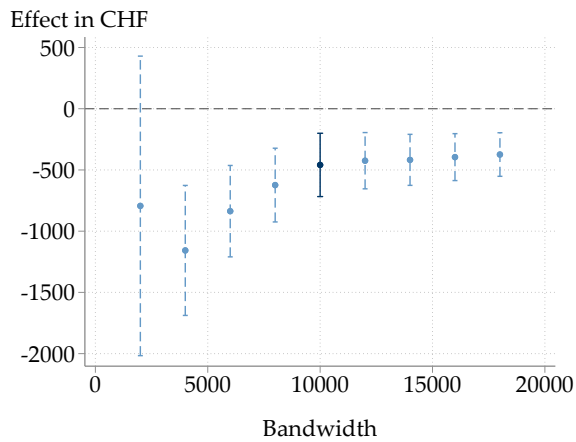
Source: Author's calculations based on administrative tax data from the canton of Bern.

FIGURE A.8: SENSITIVITY OF RD ESTIMATES TO BANDWIDTH

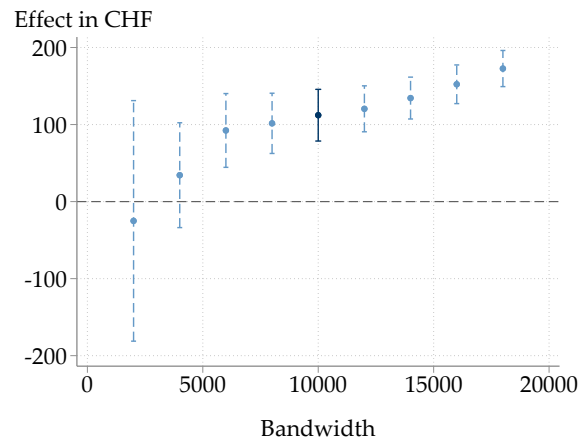
(A) TOTAL SAVINGS



(B) PRIVATE SAVINGS



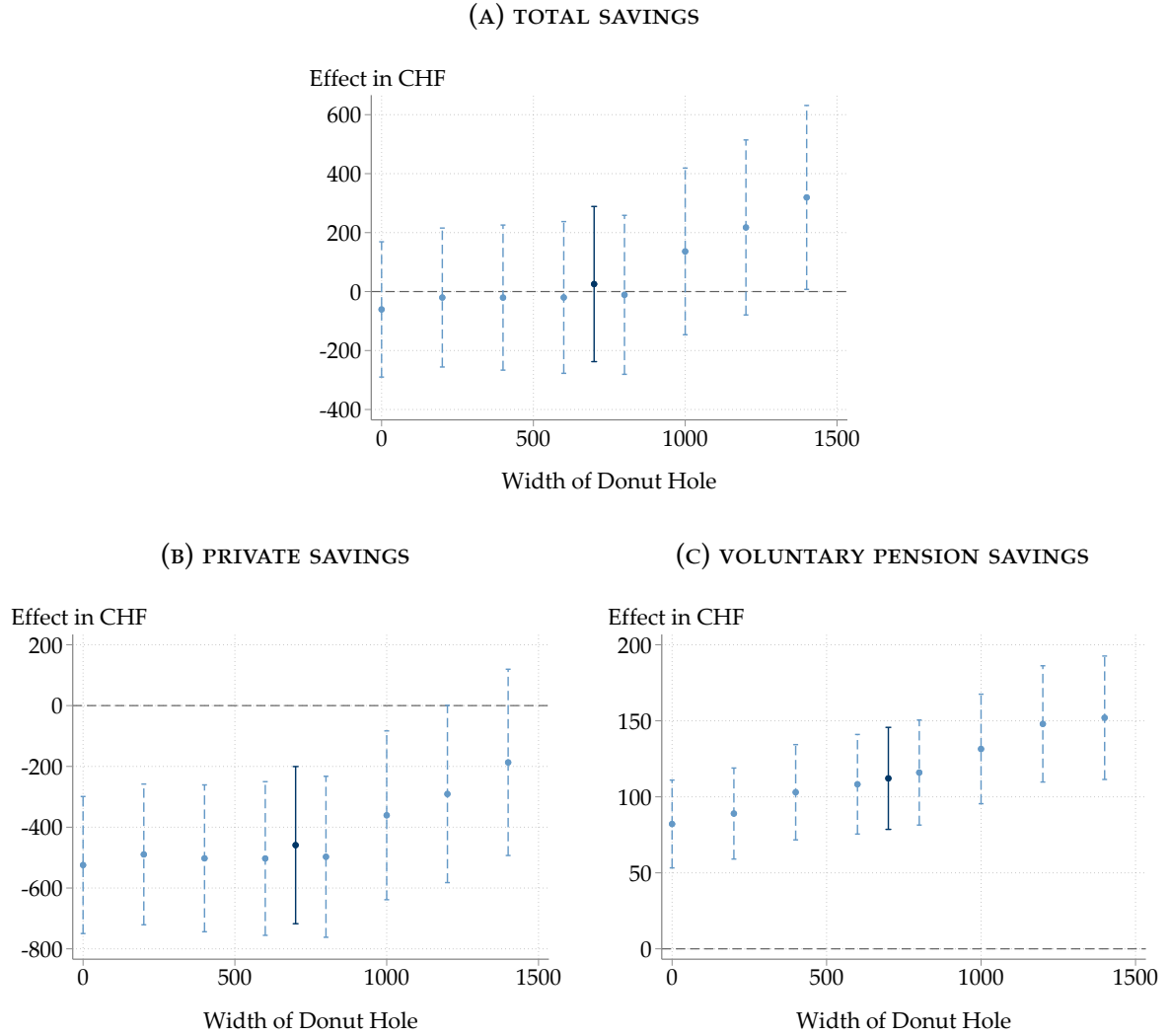
(C) VOLUNTARY PENSION SAVINGS



Notes: Coefficient plots showing the effects of the pension plan mandate depending on the bandwidth. The estimation sample is based on data from 2005 to 2017. Point estimates and 95% confidence intervals are obtained from estimating Equation (2) using a triangular kernel. The dark marker represents the main result using a bandwidth of CHF 10,000; the dashed light markers represent estimates using alternative bandwidths. Overall private savings are winsorized at percentiles 5 and 95. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

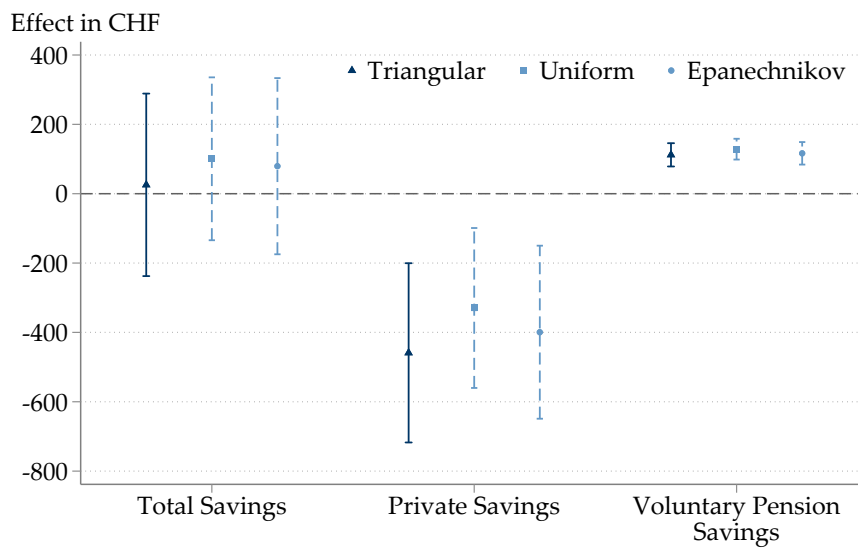
FIGURE A.9: SENSITIVITY OF RD ESTIMATES TO SIZE OF DONUT HOLE



Notes: Coefficient plots showing the effects of the pension plan mandate depending on the size of the donut hole removed from the estimation sample. The estimation sample is based on data from 2005 to 2017. Point estimates and 95% confidence intervals are obtained from estimating Equation (2) using a triangular kernel. The dark marker represents the main result using a donut hole of CHF 700; the dashed light markers represent estimates using alternative donut hole sizes. Overall private savings are winsorized at percentiles 5 and 95. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

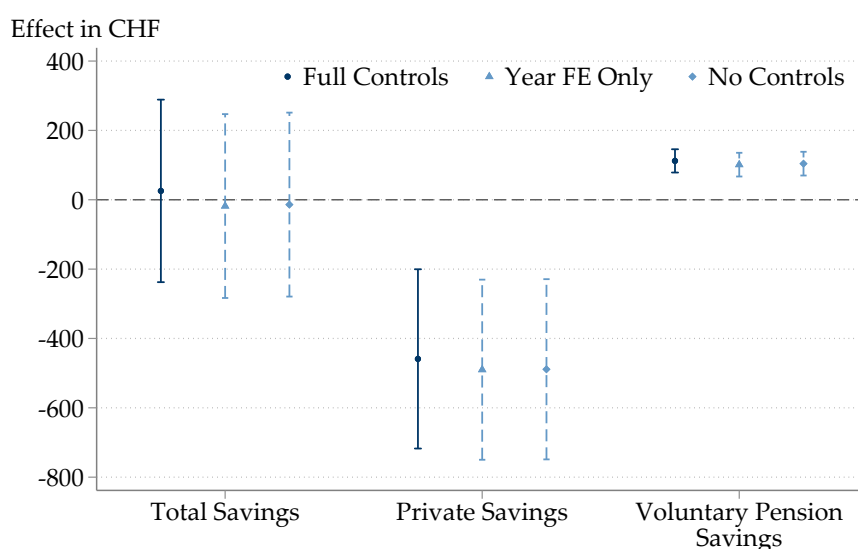
FIGURE A.10: SENSITIVITY OF RD ESTIMATES TO KERNEL



Notes: Coefficient plot showing the effects of the pension plan mandate depending on the kernel weights. Effective estimation sample includes 597,120 worker-year observations from 2005 to 2017. Point estimates and 95% confidence intervals are obtained from estimating Equation (2). The dark marker represents the main result using a triangular kernel; the dashed light markers represent estimates using alternative kernels. Overall private savings are winsorized at percentiles 5 and 95. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

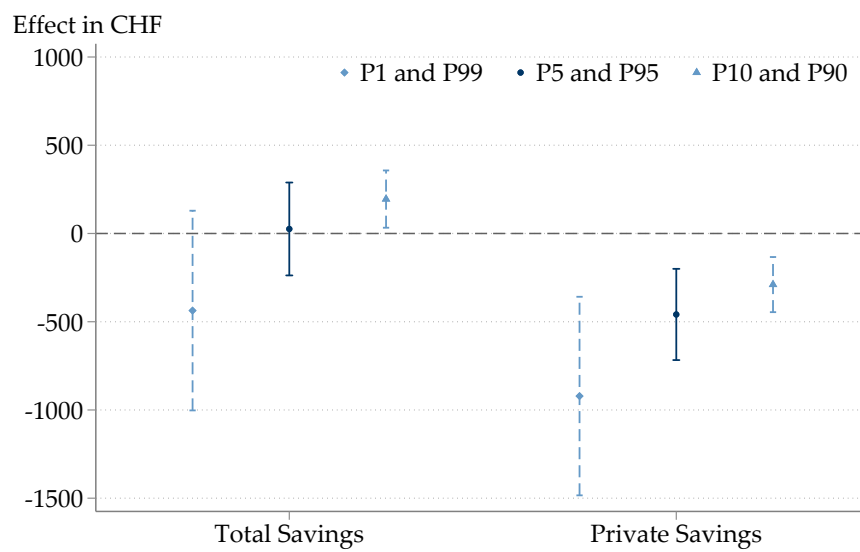
FIGURE A.11: SENSITIVITY OF RD ESTIMATES TO CONTROL VARIABLES



Notes: Coefficient plot showing the effects of the pension plan mandate depending on the set of controls included. Effective estimation sample includes 597,120 worker-year observations from 2005 to 2017. Point estimates and 95% confidence intervals are obtained from estimating Equation (2). The dark marker represents the main result estimated with the full set of controls including year fixed effects, age fixed effects, gender, and marital status; the dashed light markers represent estimates including only year fixed effects or no controls at all. Overall private savings are winsorized at percentiles 5 and 95. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

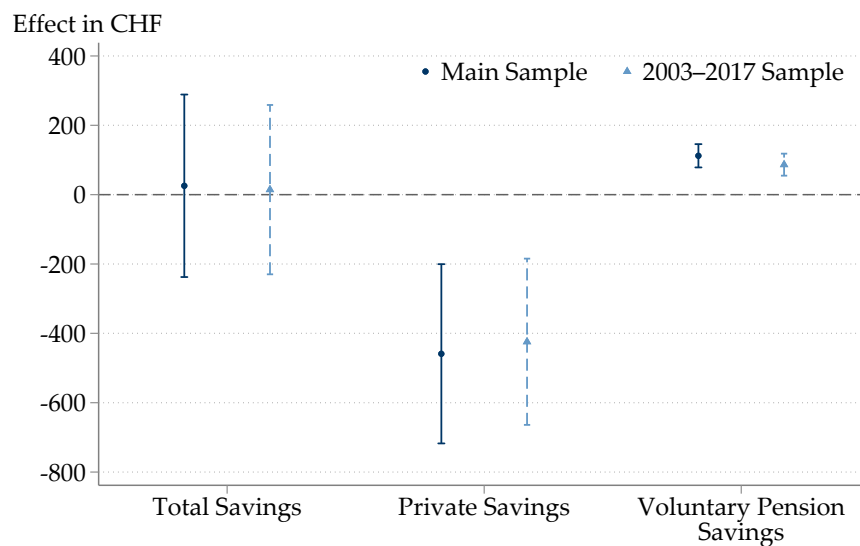
FIGURE A.12: SENSITIVITY OF RD ESTIMATES TO WINSORIZATION



Notes: Coefficient plot showing the effects of the pension plan mandate depending on the winsorization cutoffs. Effective estimation sample includes 597,120 worker-year observations from 2005 to 2017. Point estimates and 95% confidence intervals are obtained from estimating Equation (2). The dark marker represents the main result based on winsorizing private savings at percentiles 5 and 95; the dashed light markers represent estimates using private savings winsorized at percentiles 1 and 99 or percentiles 10 and 90, respectively. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

FIGURE A.13: SENSITIVITY OF RD ESTIMATES TO SAMPLE PERIOD



Notes: Coefficient plot showing the effects of the pension plan mandate depending on the sample period. Point estimates and 95% confidence intervals are obtained from estimating Equation (2). The dark marker represents the main result using data from years 2005–2017; the dashed light marker represents estimates using data from the full period 2003–2017. Overall private savings are winsorized at percentiles 5 and 95. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions and estimation.

Source: Author's calculations based on administrative tax data from the canton of Bern.

TABLE A.1: SUMMARY STATISTICS

	Mean	Median	Std. Dev.
Panel A: Demographics			
Age	42.61	43	10.22
Female	0.82	1	0.39
Married	0.67	1	0.47
Number of Children	0.93	1	1.07
Panel B: Income			
Household Income	79,783	79,585	70,182
Total Individual Income	25,343	24,123	31,176
Gross Earnings Main Job	20,689	21,146	6,013
Gross Earnings Side Job	734	0	3,043
Self-Employment Income	684	0	5,498
Other Income	3,237	59	30,056
Panel C: Wealth (Stock Variables)			
Net Wealth	60,722	17,648	102,707
Financial Wealth	42,661	15,054	63,191
Business Wealth	488	0	1,902
Property Wealth	87,469	0	122,001
Other Wealth	2,312	0	4,525
Debt	-83,318	-2,000	116,742
Panel D: Savings (Annual Flows)			
Total Savings	4,623	1,151	20,830
Mandatory Pension Savings	245	232	276
Voluntary Pension Savings	1,203	0	2,821
Private Pension Savings	1,132	0	2,113
Occupational Pension Buy-ins	72	0	1,706
Private Savings	3,175	313	20,361
Financial Savings	1,997	0	13,754
Business Savings	78	0	1,595
Property Savings	3,397	0	16,644
Other Savings	23	0	2,334
Debt Savings	-683	0	6,643
Number of Observations	597,120		
Number of Individuals	191,400		

Notes: Mean, median, and standard deviation of demographic, income, wealth, and savings variables in the estimation sample. The sample includes workers aged 25–60 with earnings within CHF 10,000 of the mandate threshold in the period from 2005 to 2017. All monetary values are reported in Swiss francs. All wealth variables and all private savings variables are winsorized at percentiles 5 and 95, except business savings and property savings which are winsorized at percentiles 2.5 and 97.5. See Section 3 for more information on data preparation and variable construction as well as Section 4.2 for more detail on sample restrictions.

Source: Author's calculations based on administrative tax data from the canton of Bern.

B The Swiss Old-Age Provision System

This appendix provides a more comprehensive description of the Swiss old-age provision system, focusing on years 2002 to 2017, which is the period covered by the tax records.³ Similar to other countries, the Swiss old-age provision system has three pillars: (i) a pay-as-you-go system (“old-age insurance”), (ii) an occupational pension system that is compulsory above an earnings threshold, and (iii) voluntary private pension accounts with a contribution cap. Appendix Table B.1 reports year-specific information on key parameters of the old-age provision system since the introduction of the occupational pension system in 1985.

B.1 Old-Age Insurance

Old-age insurance is organized as a pay-as-you-go scheme and compulsory for all individuals living or working in Switzerland between the age of 18 years and the statutory retirement age which is 64 years for women and 65 years for men. Between 2002 and 2017, the contribution rate applied to gross earnings was constant at 8.4%, of which employer and employees each pay half. The contribution of employees are deducted from their earnings by the employer and directly transferred to the social insurance administration. There is no cap on contributions, even if they do not lead to higher benefits. While benefit levels do depend on average earnings and the number of contribution years, the maximum benefit level is only double the minimum benefit level. From 2002 to 2017, the minimum annual benefit level was gradually increased from CHF 12,360 to CHF 14,100, while the maximum benefit level, accordingly, was rising from CHF 24,720 to CHF 28,200. This is relevant because key parameters of the occupational pension system are defined with respect to the maximum pension from old-age insurance as I explain in more detail in the next section. Benefit levels are usually adjusted every other year based on the evolution of an index reflecting the arithmetic mean of nominal wage growth and inflation.

B.2 Occupational Pension System

In contrast to old-age insurance, the occupational pension system is fully funded. Female employees between 25 and 64 years of age as well as male employees between 25 and 65 years of age must be enrolled in an occupational pension plan by their employer if the annual gross earnings in their main job with the same employer

³For more information on the Swiss old-age provision system and relevant changes over time, see the website of the Federal Social Insurance Office: <https://www.bsv.admin.ch/bsv/en/home/social-insurance.html> [accessed on 20 October 2021].

exceed the threshold specified in the law. Employees cannot opt out of contributing to an occupational pension account if they meet the legal criteria.

Occupational pension funds are a key tool for wealth accumulation in Switzerland. Savings in occupational pension accounts total around 8% of GDP per year. Wealth accumulated in those accounts reached almost CHF 900 billion or 130% of GDP in 2017 (Federal Statistical Office, 2019).⁴ According to data published by the Swiss National Bank, insurance and pension schemes – which mainly consist of capital in occupational pension funds – accounted for about 23% of total household wealth in Switzerland in 2020 (Annaheim and Heim, 2021).

The mandate threshold determining enrollment into occupational pension plans provides the identifying variation for the empirical strategy employed in the paper. From 1985 until 2004, the threshold had been equal to the maximum pension from old-age insurance in order to avoid overinsurance of the salary already covered by old-age insurance. As part of a multi-step reform of the occupational pension system implemented between 2004 and 2006, the threshold was lowered to 3/4 of the maximum benefit level in 2005, resulting in a drop of the threshold from CHF 25,320 in 2004 to CHF 19,350 in 2005. Before and after the reform, the threshold has been gradually increased in proportion to the rising maximum old-age insurance benefit level.⁵ Note that the mandate only applies to earnings in the main job.⁶ Employees with multiple jobs who exceed the mandate threshold only when summing up multiple salaries can join an occupational pension fund on a voluntary basis. Yet, the number of individuals making use of that option seems to be negligible (Ecoplan, 2010).⁷ Self-employed individuals can also voluntarily join an occupational pension plan in which case they are subject to the same cap on tax-deductible private pension contributions as employees covered by the mandate (see Appendix Section B.3).

To compute mandatory occupational pension contributions, a worker's qualifying earnings are multiplied by an age-specific contribution rate.⁸ Qualifying earnings are equal to gross earnings minus a deduction. Similar to the mandate threshold, the deduction's purpose is to prevent overinsurance. It had been equal to the threshold between 1985 and 2004, but in the 2005 reform the threshold was reduced by more than the deduction. Since then, the deduction equals 7/8 of the maximum benefit

⁴Switzerland's GDP in 2017 was CHF 694 billion according to data from the State Secretariat for Economic Affairs: <https://www.seco.admin.ch/seco/en/home/wirtschaftslage---wirtschaftspolitik/Wirtschaftslage/bip-quartalsschaetzungen-/daten.html> [accessed on 19 October 2021].

⁵Figure B.1 displays the evolution of the earnings threshold since 1985.

⁶If employees are not working for the same employer for the whole year, the mandate threshold applies to the hypothetical earnings they would have received if they had worked at that salary for the full year. However, employees must be on a permanent contract or work for the same employer for at least three months. I cannot distinguish these individuals from those who are not subject to the mandate because the tax data only contain information on annual earnings (see Appendix Section C.2).

⁷This finding holds up in more recent data, see Schöchli (2021).

⁸See Equation (4) in Appendix Section C.4 for the computation of mandatory pension savings.

from old-age insurance while the threshold is $3/4$, as mentioned earlier. The deduction ranges between CHF 22,575 and CHF 25,320 during the sample period.

The statutory contribution rate is 7% for employees aged 25–34, 10% for those aged 35–44, 15% for those aged 45–54, and 18% for women aged 55–64 and men aged 55–65.⁹ Employers are legally obliged to pay at least 50% of the contribution. Occupational pension funds have some autonomy in designing their plans if they want to go beyond the minimum standards defined in the law (Dorn and Sousa-Poza, 2003).

For employees whose earnings are only slightly higher than the mandate threshold, there are minimum qualifying earnings equivalent to $1/8$ of the maximum old-age insurance pension, varying between CHF 3,090 in 2002 and CHF 3,525 in 2017. Their contributions are calculated by applying the contribution rate to minimum qualifying earnings, not just to the marginal earnings above the threshold or deduction. Hence, there is a discontinuity in mandatory pension savings at the cutoff.

If employees' earnings exceed an upper bound, there are maximum qualifying earnings equal to the upper bound minus the deduction. The upper bound is defined as three times the maximum pension from old-age insurance, gradually increasing from CHF 74,160 in 2002 to CHF 84,600 in 2017.¹⁰ Furthermore, many occupational pension plans allow one-off “buy-ins” that individuals can voluntarily make beyond the mandatory contributions. These can also be deducted from taxable income, but they are only allowed up to the level of savings that a worker would have accumulated if they had been receiving their current salary since they were 25 years old (Kuhn, 2020).

To illustrate the calculation of contributions, Appendix Figure A.1 displays the relationship between mandatory occupational pension contributions and gross earnings for an employee aged 45–54 (contribution rate of 15%) in 2017. The discontinuity in mandatory pension savings at the cutoff is clearly visible. This relationship follows the same pattern in all other years with slight differences based on the year-specific parameters listed in Appendix Table B.1.

Retirees can choose to receive occupational pension benefits as some combination of a lifelong annuity and a lump-sum payment. By law, they are entitled to receive at least a quarter of their occupational pension capital as a lump sum.

The transformation of pension capital into an annuity is subject to a legally de-

⁹Women's legal retirement age in the occupational pension system was raised from 62 to 64 years in 2005 in order to be aligned with the rules of old-age insurance. Because of the lower retirement age, the statutory contribution rates applied to slightly different age groups for women before 2005 – specifically, 7% for the age group 25–31 years, 10% for the age group 32–41 years, 15% for the age group 42–51 years, and 18% for the age group 52–62 years.

¹⁰Most occupational pension funds provide insurance for the portion of the salary above the upper bound on a voluntary basis (Bütler, 2009). This practice is not relevant for employees in the earnings range studied in this paper.

defined minimum conversion rate. The minimum conversion rate is only binding for the calculation of benefits based on pension capital deriving from mandatory savings. It does not apply to voluntary occupational pension savings. The rate was reduced gradually from 7.2% to 6.8% over the sample period as a consequence of the reform implemented between 2004 and 2006. The conversion rate translates the occupational pension capital into annual benefit entitlements. Accordingly, CHF 100,000 are converted into an annuity of CHF 6,800–7,200 depending on the year-specific conversion rate. Old-age insurance and occupational pension benefits are designed to jointly replace around 60% of pre-retirement earnings.

The tax treatment of occupational pension savings is quite favorable. Contributions and investment returns are exempt from income tax (and there is no capital gains tax in Switzerland). Moreover, pension capital is exempt from wealth tax. Benefits are subject to tax: while standard income tax is levied on annuities, lump-sum receipts are taxed at special, rather advantageous rates.

B.3 Private Pension Savings

The third pillar consists of voluntary contributions to designated private pension accounts. Because these savings can be deducted from taxable income, they are subject to a contribution cap. Employees enrolled in an occupational pension plan are allowed to make annual contributions to private pension accounts of up to 8% of the upper bound of qualifying earnings in the second pillar (equivalent to 24% of the maximum old-age insurance pension). This cap has gradually increased from CHF 5,933 in 2002 to CHF 6,768 in 2017. Self-employed individuals who voluntarily join an occupational pension plan are subject to the same cap on private pension savings. Individuals who are not enrolled in an occupational pension plan, whether employed or self-employed, can contribute up to 20% of their net earnings but at most 40% of the upper bound of qualifying earnings in the occupational pension system (ranging between CHF 29,664 in 2002 and CHF 33,840 in 2017) to private pension accounts.

Private pension accounts are set up independently from occupational pension accounts, either at a bank or insurance company. The law allows contributions to standard savings accounts as well as investments into stocks and other securities. Use of this savings vehicle is fairly widespread: private pension capital in designated accounts totalled CHF 121 billion or 17.4% of GDP in 2017.¹¹

Until people enter retirement, they have limited access to the accumulated capital in these designated private pension accounts. Once eligible, disbursement can be in

¹¹Statistics on the occupational pension and private pension system are available from the Federal Social Insurance Office: <https://www.bsv.admin.ch/bsv/en/home/social-insurance/bv/statistik.html> [accessed on 20 October 2021].

the form of an annuity or lump sum.

The tax treatment of private pension savings is similar to occupational pension savings: contributions can be deducted from taxable income, capital is exempt from wealth tax, while annuities are subject to income tax and lump-sum payments are taxed at preferential rates at the time of receipt.

B.4 Reform of the Occupational Pension System in 2004–2006

The 2004–2006 reform was the first major reform of the occupational pension system since its introduction in 1985, and it remains the only one to date.¹² It was implemented in three steps between 2004 and 2006. Before the reform, from 1985 to 2004, both the earnings threshold and the deduction for computing mandatory occupational pension savings had been equal to the maximum pension in the pay-as-you-go system. Accordingly, they had been increasing one-for-one with the rising maximum benefit level during that period, reaching CHF 25,320 in 2004.

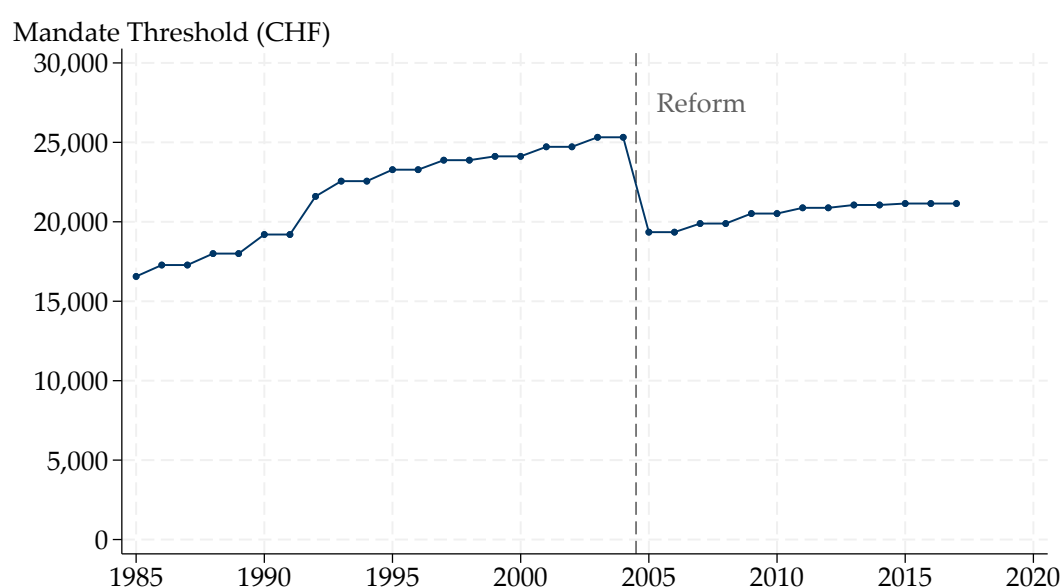
As part of the reform, both the threshold and the deduction were lowered in 2005. Appendix Figure B.1 plots the evolution of the threshold since introduction of the occupational pension system in 1985, demonstrating the significance of the 2005 reduction. The 2005 reform reduced the threshold to 3/4 of the maximum pay-as-you-go pension. As a consequence, it fell from CHF 25,320 in 2004 to CHF 19,350 in 2005. Around 140,000 employees in Switzerland who otherwise would not have been subject to the mandate became covered due to the reform, the majority of which were female, working part-time, and had low hourly wages below CHF 25 (Ecoplan, 2010). This represents an increase in the number of workers covered by roughly 4%. The stated purpose of the reduction of the threshold was to improve financial preparedness for retirement among low-income workers.

The deduction was lowered to 7/8 of the maximum pay-as-you-go pension, i.e., to CHF 22,575 in 2005. This increased mandatory occupational pension savings at almost all earnings levels above the threshold. Lowering the deduction aimed to offset some of the decline in pension benefit levels caused by the decrease in the conversion rate implemented in the same reform.

The reform included further policy changes such as lowering the statutory rate for converting pension account balances into annuities from 7.2% to 6.8% over a transition period of ten years, increasing women's retirement age in the occupational pension system to 64 years in line with the rules in the pay-as-you-go system, and aligning the contribution rate schedule for women with that of men.

¹²Detailed information on the reform ("1. BVG-Revision") is available from the Federal Social Insurance Office: <https://www.bsv.admin.ch/bsv/de/home/sozialversicherungen/bv/reformen-und-revisionen/revision-1-bvg.html> [accessed on 30 October 2021].

FIGURE B.1: EARNINGS THRESHOLD OF THE OCCUPATIONAL PENSION PLAN MANDATE



Notes: The figure displays the evolution of the earnings threshold determining coverage of the pension plan mandate since the introduction of the occupational pension system in 1985. The dashed vertical line indicates the timing of the reform that lowered the threshold in 2005. Employees with gross earnings in their main job exceeding the threshold are subject to the mandate.

Source: Author's illustration based on information from the Federal Social Insurance Office.

TABLE B.1: KEY PARAMETERS OF THE OLD-AGE PROVISION SYSTEM IN SWITZERLAND

Year	Old-age insurance		Occupational pension system				Private pension savings
	Minimum benefit	Maximum benefit	Threshold	Deduction	Upper bound	Min. qual. earnings	Contribution cap
1985	8,280	16,560	16,560	16,560	49,680	2,070	-
1986	8,640	17,280	17,280	17,280	51,840	2,160	-
1987	8,640	17,280	17,280	17,280	51,840	2,160	4,147
1988	9,000	18,000	18,000	18,000	54,000	2,250	4,320
1989	9,000	18,000	18,000	18,000	54,000	2,250	4,320
1990	9,600	19,200	19,200	19,200	57,600	2,400	4,608
1991	9,600	19,200	19,200	19,200	57,600	2,400	4,608
1992	10,800	21,600	21,600	21,600	64,800	2,700	5,184
1993	11,280	22,560	22,560	22,560	67,680	2,820	5,414
1994	11,280	22,560	22,560	22,560	67,680	2,820	5,414
1995	11,640	23,280	23,280	23,280	69,840	2,910	5,587
1996	11,640	23,280	23,280	23,280	69,840	2,910	5,587
1997	11,940	23,880	23,880	23,880	71,640	2,985	5,731
1998	11,940	23,880	23,880	23,880	71,640	2,985	5,731
1999	12,060	24,120	24,120	24,120	72,360	3,015	5,789
2000	12,060	24,120	24,120	24,120	72,360	3,015	5,789
2001	12,360	24,720	24,720	24,720	74,160	3,090	5,933
2002	12,360	24,720	24,720	24,720	74,160	3,090	5,933
2003	12,660	25,320	25,320	25,320	75,960	3,165	6,077
2004	12,660	25,320	25,320	25,320	75,960	3,165	6,077
2005	12,900	25,800	19,350	22,575	77,400	3,225	6,192
2006	12,900	25,800	19,350	22,575	77,400	3,225	6,192
2007	13,260	26,520	19,890	23,205	79,560	3,315	6,365
2008	13,260	26,520	19,890	23,205	79,560	3,315	6,365
2009	13,680	27,360	20,520	23,940	82,080	3,420	6,566
2010	13,680	27,360	20,520	23,940	82,080	3,420	6,566
2011	13,920	27,840	20,880	24,360	83,520	3,480	6,682
2012	13,920	27,840	20,880	24,360	83,520	3,480	6,682
2013	14,040	28,080	21,060	24,570	84,240	3,510	6,739
2014	14,040	28,080	21,060	24,570	84,240	3,510	6,739
2015	14,100	28,200	21,150	24,675	84,600	3,525	6,768
2016	14,100	28,200	21,150	24,675	84,600	3,525	6,768
2017	14,100	28,200	21,150	24,675	84,600	3,525	6,768

Notes: Year-specific parameters of the old-age provision system in Switzerland since the introduction of the occupational pension system in 1985. All values are reported in Swiss francs. The option to contribute to preferentially taxed private pension accounts was established in 1987. The cap on these private pension contributions applies to individuals enrolled in an occupational pension plan which includes all employees with earnings above the mandate threshold.

Source: Official information from the Federal Social Insurance Office.

C Data Appendix

In this appendix, I describe the administrative data provided by the tax authority of the canton of Bern in more detail. I start by discussing what variables are available at the individual or the household level, and how I split up couples into individual observations. Subsequently, I turn to the income, wealth, and savings data used in the paper. Finally, I discuss data cleaning and sample restrictions.

C.1 Individual-Level vs. Household-Level Information

In Switzerland, the relevant tax unit is the household. Thus, married couples jointly file one tax return. Because the pension plan mandate applies at the individual level, I follow the approach of [Fagereng et al. \(2020\)](#) and split up married couples into individual observations, equally assigning half of the value of income and wealth variables that are only reported at the household level to each partner. All wealth information is reported at the household level. Income variables reported at the household level include self-employment income, business income, financial income, and property income. Most important for my analysis, earnings from employment are reported at the individual level, as are transfer income and pension income. The same is true for occupational pension buy-ins and private pension savings.

C.2 Income

The pension plan mandate applies to gross earnings in the main job, before any deductions are applied. The tax records contain separate information for earnings in the main job and other jobs. The definition of earnings in the main job used by the tax administration overlaps to a large degree with the definition relevant for the mandate (see Appendix Section [B.2](#)). Both refer to total earnings with the same employer. There is a small difference in that the tax authority may include positions at multiple employers as part of the main job if they are similar in terms of working hours or income received, while the mandate applies only to the main job at one employer. Yet, this distinction is unlikely to be relevant for many employees in the data, so the resulting measurement error should be limited.

Employees report earnings net of social insurance and mandatory occupational pension contributions to the tax administration because these are deducted directly from their salary by the employer. Therefore, I impute gross earnings based on net earnings recorded in the tax data and the year-specific social insurance and occupational pension schedules. Because social insurance and occupational pension contributions vary along the earnings distribution, the calculation must differentiate

between certain earnings ranges.¹³ Taking as an example an individual subject to the pension plan mandate for whom neither the minimum nor maximum qualifying earnings of the occupational pension system are binding, the calculation must take into account that social insurance contribution rates are applied to total gross earnings and occupational pension contribution rates are applied to gross earnings above the deduction. Overall, gross earnings in the main job, G_{it} , of employee i in year t can be imputed from net earnings as

$$G_{it} = \left\{ \begin{array}{ll} \frac{N_{it}}{1-i_{it}^e} & \text{if } N_{it} < (C_t \times (1 - i_{it}^e) - p_{it}^e \times M_t) \\ \frac{N_{it}}{1-i_{it}^e} \text{ or } \frac{N_{it}+p_{it}^e \times M_t}{1-i_{it}^e} & \text{if } N_{it} \geq (C_t \times (1 - i_{it}^e) - p_{it}^e \times M_t) \text{ and } N_{it} < (C_t \times (1 - i_{it}^e)) \\ \frac{N_{it}+p_{it}^e \times M_t}{1-i_{it}^e} & \text{if } N_{it} \geq (C_t \times (1 - i_{it}^e)) \text{ and } N_{it} < ((D_t + M_t) \times (1 - i_{it}^e) - p_{it}^e \times M_t) \\ \frac{N_{it}-p_{it}^e \times D_t}{1-i_{it}^e-p_{it}^e} & \text{if } N_{it} \geq ((D_t + M_t) \times (1 - i_{it}^e) - p_{it}^e \times M_t) \text{ and } N_{it} < (B_t \times (1 - i_{it}^e) - p_{it}^e \times (B_t - D_t)) \\ \frac{N_{it}+p_{it}^e \times (B_t-D_t)}{1-i_{it}^e} & \text{if } N_{it} \geq (B_t \times (1 - i_{it}^e) - p_{it}^e \times (B_t - D_t)), \end{array} \right. \quad (3)$$

where N_{it} denotes net earnings, C_t is the cutoff value of the pension plan mandate, D_t is the deduction, M_t are the minimum qualifying earnings, and B_t is the upper bound in the occupational pension system, while i_{it}^e represents the employee share of the social insurance contribution rate, and p_{it}^e represents the employee share of the age-specific occupational pension contribution rate.¹⁴ Employees and employers each pay half of total social insurance contributions. Equally, I set the employee share of occupational pension contributions to 50% which is the maximum employee share defined in the law. Some employers may bear more than 50% (which cannot be observed in the tax data), so this may result in slight measurement error.

Note that due to the discontinuity in mandatory occupational pension contributions, gross earnings slightly below and slightly above the mandate threshold can result in the same net earnings recorded in the tax data. Thus, gross earnings cannot be unambiguously imputed from net earnings for a small number of individuals (captured by the second line in Equation (3)). This problem only concerns a narrow

¹³Note that the definition of the different earnings ranges with respect to net earnings in Equation (3) is equivalent to the definition in terms of gross earnings in Equation (4).

¹⁴Social insurance contributions include contributions for old-age insurance, disability insurance, loss of earnings compensation, and unemployment insurance. Above a high earnings cutoff, unemployment insurance contribution rates are reduced in most years that I have data on. I account for the reduced unemployment insurance contributions when calculating gross earnings but omit it from Equation (3) because it does not matter for individuals in the earnings range of interest for this paper. More information on the social insurance system in Switzerland and historical contribution schedules are available on the website of the Federal Social Insurance Office: <https://www.bsv.admin.ch/bsv/de/home/sozialversicherungen/ueberblick/beitraege.html> [accessed on 23 October 2021].

earnings range of less than CHF 350 below and above the threshold (with the exact width depending on the age-specific contribution rate and the year-specific minimum qualifying earnings). I exclude these observations from the sample throughout the paper. Only in the robustness checks setting the donut hole to less than CHF 350 (Appendix Figure A.9), I include these observations, assigning them with equal probability the corresponding gross earnings below or above the mandate threshold.

The tax records also include information on other types of income that are less relevant for this paper, such as self-employment income, business income, financial income, property income, transfer income, and pension income.

C.3 Wealth

The tax records contain detailed information on wealth by asset type, including financial wealth, business wealth, property wealth, other types of wealth, and debt. The Swiss wealth tax is quite comprehensive, covering all types of assets except for pension wealth in occupational and private pension accounts which is thus missing from the data. This is not a problem for the empirical analysis in this paper, because pension savings can be observed or imputed even without directly observing pension wealth. Information on pension contributions is sufficient because pension wealth generally cannot be accessed during the work life.

It needs to be noted that the valuation of property for tax purposes systematically underestimates the true market value. As a rule of thumb, residential property is valued at around 60% of its market value in Switzerland (OECD, 2018c), although in individual cases the valuation may deviate significantly from that benchmark. To analyze the discrepancy between tax value and market value of residential property, the tax administration of the canton of Bern conducted an analysis comparing the observed price of all housing transactions in a given year to the value of these properties in the tax records.¹⁵ On average, the tax value of property was 71% of its market value in 2002. Because there was no revaluation during the sample period and property prices in Bern have generally been increasing, the tax value has gradually declined to about 55% of the market value in 2017.

C.4 Savings

Any type of pension or private savings could be affected by the pension plan mandate. Some of these are directly observable in the tax data; others need to be imputed

¹⁵The analysis of property valuation for tax purposes is available on the website of the tax authority of the canton of Bern: https://www.sv.fin.be.ch/sv_fin/de/index/navi/index/steuersituationen/kauf-verkauf_liegenschaft/amtlicher_wert/allgemeine-neubewertung20.html [accessed on 23 October 2021].

based on information available in the dataset.

As mentioned earlier, mandatory occupational pension contributions are withheld at source by the employer, so they are not recorded in the tax data. I impute these by applying the statutory contribution rates – the sum of the employer and employee shares – to individuals' gross earnings in the main job. Based on the contribution schedule explained in Section 2.1, mandatory occupational pension savings, S_{it}^{occ} , of employee i in year t are calculated as

$$S_{it}^{occ} = \begin{cases} 0 & \text{if } G_{it} < C_t \\ p_{it} \times M_t & \text{if } G_{it} \geq C_t \quad \text{and } (G_{it} - D_t) < M_t \\ p_{it} \times (G_{it} - D_t) & \text{if } (G_{it} - D_t) \geq M_t \quad \text{and } G_{it} < B_t \\ p_{it} \times (B_t - D_t) & \text{if } G_{it} \geq B_t, \end{cases} \quad (4)$$

where p_{it} is the age-specific total mandatory pension contribution rate, i.e., employer and employee share. All other variables are defined as in Equation (3). Appendix Figure A.1 illustrates how mandatory occupational pension savings are computed from gross earnings.

Private pension savings in preferentially taxed accounts as well as voluntary occupational pension buy-ins are directly observed at the individual level in the tax data, because they must be reported in the tax return in order to be deductible from taxable income.

I compute overall private savings as the change in net wealth relative to the previous year. Net wealth represents the difference between gross wealth and debt, and can be directly observed in the tax records. Since mandatory and voluntary pension savings are tax-deductible and thus measured in pre-tax terms, I compute pre-tax private savings by dividing observed post-tax private savings by the individual's net-of-marginal-tax rate on income. Therefore, private savings, S_{it}^{priv} , of individual i in year t are given by

$$S_{it}^{priv} = \frac{W_{i,t} - W_{i,t-1}}{1 - \tau_{it}}, \quad (5)$$

where $W_{i,t}$ denotes net wealth of individual i in year t and τ_{it} denotes their marginal income tax rate. Because the income tax schedule in the canton of Bern is smooth, with marginal tax rates continuously changing across the distribution, I estimate marginal tax rates by fitting a global fifth-order polynomial regression of the income tax liability on total household income. This is done separately for single and married individuals because they face different tax schedules. To reduce the influence of outliers, I focus on a window of household income between CHF 10,000 and CHF 200,000. Marginal tax rates are set to zero for individuals with incomes below CHF 10,000 and capped at the rate estimated for CHF 200,000 for those above that threshold.

This measure of private savings corresponds to *gross savings* as it includes accrued capital gains from changes in asset prices (which are sometimes called “passive savings” in the literature) (see [Fagereng et al., 2025](#)). Note that it does not capture accrued gains on property because there has not been a revaluation of property during the observed period. I cannot separate gross savings into *net savings* (sometimes called “active savings”) and capital gains because realized gains are not observed as Switzerland does not have a capital gains tax and I do not have data down to the level of single assets or transactions.¹⁶ An implication of this is that private savings are highly variable due to fluctuations in asset prices. Another reason for the variability of private savings is the purchasing timing of durable goods, services, and lumpy non-durables ([Chetty et al., 2014](#)).

Total savings are defined as the sum of all savings variables discussed above, including mandatory occupational pension savings, occupational pension buy-ins, private pension savings, and private savings. All of them are measured in pre-tax terms.

C.5 Data Cleaning and Sample Restrictions

To prepare the data for the empirical analysis, I remove observations that are unreliable or not comparable to standard taxpayers (similar to [Brunner, Meier and Näf, 2020](#)). This group includes individuals who are taxed for only part of the year because they move abroad or arrive from abroad (1.9% of all observations), individuals who fail to hand in a tax return and are assessed by the tax authority (2.8%), duplicate observations for individuals in the same year (0.6%), and observations with obvious errors in the reported information (0.1%).

¹⁶The latter is a common challenge in the literature on the measurement of savings. A few papers leverage very detailed administrative data to distinguish net savings and capital gains, including [Bach, Calvet and Sodini \(2020\)](#), [Fagereng et al. \(2020\)](#), and [Fagereng et al. \(2025\)](#). [Fagereng et al. \(2025\)](#) provide a helpful discussion of the distinction between gross and net savings and its implications, using both economic theory and an empirical application drawing on Norwegian data.